

Figure 7-18 Change in bone density of the contralateral ulna following ostectomy of a radius. Compared with the situation in Figure 7-19, there is little change in bone density because both the intact radius and ulna continue to share loading.

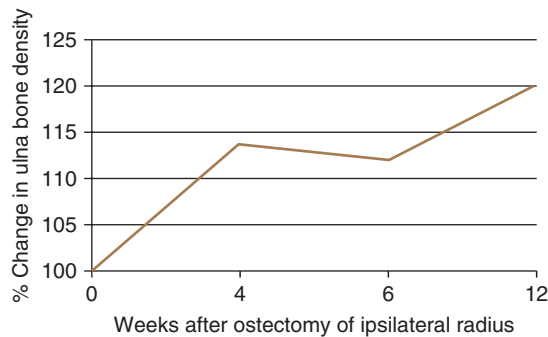
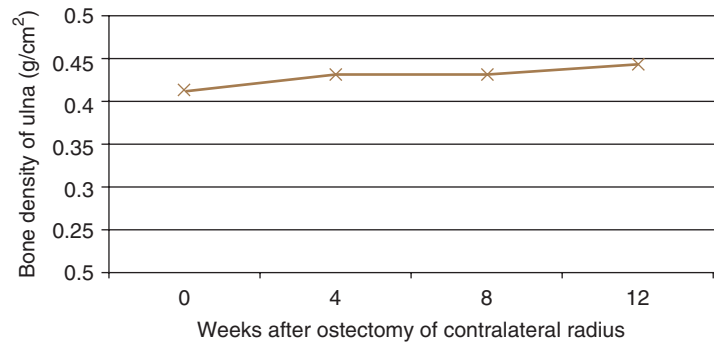


Figure 7-19 Relative change of the bone density of the ulna following ostectomy of the ipsilateral radius. Note the increased density as a result of the shift of weight bearing from a paired two-bone system to the single remaining bone.

radius and ulna (Figure 7-18).³⁶⁵ This model allows weight bearing on the limb undergoing ostectomy of the radius because the intact ulna allows for weight bearing. However, because all of the weight is borne by the intact ulna following ostectomy, the load on that bone is significantly greater as compared with the load-sharing seen with the intact radius and ulna of the contralateral limb. The increased load results in significant increases in bone mass and density in the ipsilateral ulna within a relatively short period (Figure 7-19).

Increasing weight bearing on bones by exercising dogs with lead-weighted jackets results in increased bone mass.³⁶⁶ Increasing the load carried to 130% of body weight and exercising on a treadmill for 75 minutes at 3.3 km per hour during a 70-week period resulted in a mild increase in BMC of the tibia when adjusted for actual body weight. This form of activity may increase bone mass and help reduce bone loss during aging.

The effects of long distance running training on bone growth have been studied in young, growing dogs.²²⁴ Young beagle dogs trained on a treadmill for 1 year, gradually increasing running to 40 km/day at a speed of 5.5-6.8 km/hour with a 15-degree inclination during the last 15 weeks, while control dogs were kept in their cages. Ulna and radius bone mass as a ratio to body mass increased approximately 10% in trained dogs as compared

with controls, although there was no change in mass of cranial tibial and semitendinosus muscles. Long-distance running apparently has a mild positive effect on bone mass in growing dogs.

In contrast, another study of immature dogs with a similar training regimen indicated that, although intensive endurance training may increase bone mass of weight-bearing bones, the appendicular skeleton may have reduced bone density.³⁶⁷ Trained dogs had larger and heavier radius, ulna, and hip bones, but reduced BMD of thoracic vertebrae as compared with untrained controls. In addition, trained dogs developed more osteophytes in the spines but not in the extremities. Intense aerobic training of young dogs may alter bone mass, possibly because of altered biomechanical forces placed on the bones.

High-level activity, such as racing, may affect bone to a greater extent as compared with bone changes occurring with endurance training. One study of racing Greyhounds found that the cross-sectional and cortical areas of left metacarpals IV and V were greater than the right metacarpals.³⁶⁸ In addition, osteonal density was greater on the dorsal surface of the bones than the medial and lateral quadrants. These changes indicate site-specific adaptations to the bones as a result of asymmetric loading from running in a counterclockwise fashion and greater loading of the inside limb.

A lifetime of weight-bearing physical exercise may decrease the amount of bone lost through the normal aging process in dogs.³⁶⁹ Dogs in a trained group exercised for 90 minutes, five days/week, for 527 weeks, running an average of 3.3 km/hr on a treadmill while carrying approximately 11.5 kg. The control group had normal cage activity in dog runs. Long-term exercise significantly reduced cortical porosity and marginally increased cortical area in aged dogs, because of a decrease in bone resorption rather than an increase in bone formation. The results of this study have implications for the role of exercise in delaying bone loss during aging. Training appears to alter mineral uptake and retention by bone. For example, in one study of geriatric dogs exercising 40 min/d at 2.4 km/hr on a 15-degree grade, dogs had less calcium excretion on exercise days.³⁷⁰ Similar studies of humans support these

findings and also indicate that bone deposition rate is increased with training.³⁶

Although mechanical stimulation is important in promoting bone formation in normal dogs, fracture healing may also be enhanced with weight bearing. Dogs with surgically created defects in the tibial cortex that were allowed to bear weight on the limb had significantly more woven bone in the defects as compared with dogs that were not permitted to bear weight.³⁷¹ This suggests that weight bearing is a permissive factor for bone formation in bony defects. In addition, training may enhance healing if a fracture should occur after initiating training. A study of mice undergoing training prior to fracture had higher rates of collagen synthesis and higher concentrations of calcium in the healing fracture callus as compared with sedentary rats.³⁷²

Although exercise is generally thought to have favorable effects on bone health, it has also been suggested that exercise may reduce fracture risk.³⁷³ Cross-sectional studies and interventional studies using assessments such as BMD and falls support this belief. For rehabilitation programs to improve bone health, they should include not just weight-supported activities (e.g., cycling) but also weight-bearing activities (e.g., walking, resistance exercise).³⁷³

Effects of Medications on Bone

A number of studies have evaluated the effects of various compounds on bone mass in dogs. Some of these substances are osteoprotective, whereas others, such as corticosteroids, may hasten bone loss in immobilized limbs.

Some medications and hormones help protect against bone loss during immobilization. Dogs receiving 60 to 240 ng/kg of vitamin D₃ for 4 weeks had trabecular bone volume of the calcaneus in direct relation to the dosage of vitamin D₃ in a plaster cast disuse osteoporosis model for 4 weeks, with the highest dose resulting in bone volume equal to the control limb.³⁷⁴ Unfortunately, this dose is considered to be toxic. Vitamin D₃ administered to beagle dogs after ovariectomy was effective in maintaining normal urinary hydroxyproline excretion, suggesting that the treatment was effective in helping prevent the bone loss associated with ovariectomy.³⁷⁵ In fact, BMD of the lumbar vertebrae and femur of ovariectomized dogs receiving 2 or 10 mcg/kg of vitamin D₃ were the same as non-ovariectomized control animals 30 months after surgery, whereas density was 25% lower in untreated ovariectomized dogs.

Bone loss during disuse has been prevented with bisphosphonate compounds in humans following paralysis or bed rest. Using a forelimb immobilization model in dogs, pamidronate, a bisphosphonate compound, was administered for 7 days followed by 3 weeks without treatment. This cycle was repeated three times for a total of 12 weeks. This treatment was effective in preventing cancellous bone loss and maintaining cortical bone density and

stiffness.³²⁶ In addition, indices of bone formation and resorption were significantly reduced.

In another study, risedronate, also a potent inhibitor of osteoclastic activity, was evaluated to determine if it could effectively prevent cancellous bone loss in middle-aged female beagles subjected to single forelimb immobilization for 12 months.³⁷⁶ Age-matched, nonimmobilized dogs served as controls. Half the animals from each group received risedronate. Immobilization caused a dramatic reduction in cancellous bone mass (−71%) of distal second metacarpals, characterized by decreases in trabecular width (−51%) and number (−41%), and fourfold increases in indices of bone resorption. Bone formation indices were also significantly higher in immobilized dogs. Risedronate treatment reduced, but did not abolish, cancellous bone loss caused by immobilization. Immobilized animals treated with risedronate lost nearly 50% of cancellous bone mass, whereas trabecular width and number were reduced by 31% and 25%, respectively. Also, overall bone formation parameters remained roughly at intact control levels; however, mineral apposition rate relative to intact control was reduced in treated animals, indicating that high-dose risedronate treatment might suppress osteoblastic function, especially with long-term disuse. Risedronate treatment was partially effective in preventing cancellous bone loss during long-term disuse. Although bisphosphonates can impair the ability of mature osteoclasts to resorb bone, they do not completely overcome osteoclast recruitment caused by long-term disuse.

There may be differences between the effectiveness of bisphosphonates on osteoporosis resulting from disuse versus other forms of osteoporosis. One study evaluated whether risedronate can preserve cortical bone mass and mechanical properties during disuse for 12 months on metacarpal diaphyses in dogs.³⁷⁷ Risedronate slowed cortical thinning and partially preserved mechanical properties, but it was unable to suppress bone loss to the degree seen in other osteoporoses. Immobilization caused marked reduction in cortical area (−42%) and cortical thinning (−40%). Both bone resorption and formation were significantly elevated over control levels. Immobilization also decreased maximum load and stiffness by approximately 80% compared with controls. Risedronate reduced both periosteal bone loss and marrow cavity expansion, and also reduced cortical bone loss, partially conserving tissue mechanical properties. This modest effect contrasts with more dramatic actions of the bisphosphonate in other osteoporoses.

Bisphosphonate compounds also affect bone resorption in other species. A study evaluated tiludronate, a bisphosphonate, on biochemical biomarkers of bone metabolism and on bone density and structure in a model of disuse osteoporosis induced by cast application in horses.³⁷⁸ Horses were immobilized for 8 weeks, followed by a 4-week remobilization period and an 8-week standardized

training protocol. A rapid decrease was seen in type I collagen cross-links, an indicator of bone resorption, in horses receiving tiludronate. No significant differences between groups were seen in bone alkaline phosphatase, a marker of bone formation. The loss of BMD in the immobilized metacarpal III bone was significantly less in the tiludronate group as compared with immobilized placebo controls. Therefore this bisphosphonate significantly reduced bone resorption during immobilization, and helped prevent osteopenia in the immobilized limb.

Tamoxifen citrate administered to growing dogs with a hindlimb placed in a fiberglass cast for 28 days was moderately effective in sparing bone mass as compared with immobilized controls.³²⁷ The calculated bone mass sparing effect of tamoxifen was 24.4%. This drug is a nonsteroidal antiestrogen compound, with some ability to inhibit prostaglandin synthesis, modulate signal transduction processes that control cell growth, and induce TGF- β .

Aspirin may also be effective in diminishing bone loss during immobilization. Osteoporosis was induced in growing dogs using a cast immobilization model for 4 weeks.³³⁴ Bone PGE concentrations doubled and bone mass was reduced in untreated immobilized dogs. Dogs receiving 25 mg/kg of buffered aspirin every 8 hours had a 13% bone mass sparing effect and a 65% reduction in bone PGE. Similar bone-sparing effects have been observed with other NSAIDs, such as ibuprofen, in the treatment of osteolytic conditions such as osteomyelitis and bone metastases.

Bone morphogenetic proteins have also been used to help manage bone atrophy as a result of fracture. In one case report, recombinant human BMP-2 solution on a collagen sponge was placed along the diaphysis of an atrophic radius, which had a history of recurring fractures.³⁷⁹ Two months after rhBMP-2 treatment, new mineralized bone was present, which significantly increased the diameter of the radius and allowed the removal of the ESF.

Rehabilitation Considerations

The results of these studies are useful to help structure rehabilitation programs for musculoskeletal injuries. The injury must be protected from further damage in the early stages of healing. Acceleration of the remodeling process of healing tissues should be promoted by applying adequate stress and strain to the tissues in appropriate stages. The deleterious effects on biomechanical properties of tissues suggests that immobilization should be avoided or minimized whenever possible. Joints and limbs should be mobilized early, while simultaneously protecting the weakened tissue during healing, especially considering the length of time needed to achieve near-normal strength of tissues. Although certain tissues, such as the ligament substance, may recover rapidly following remobilization, other tissues, such as the bone-ligament complex, may take

as long as 1 year to return to normal. Early joint mobilization and limb usage must be tempered with the degree of injury and the repair performed, if any. The healing tissues must be protected from excessive force, and a progressive rehabilitation program should be used that applies low cyclic loads to promote healing and enhance remodeling. Progressive increases in the frequency and magnitude of loads applied to the healing tissues enhance healing and tissue strength. Although little is known regarding the optimal stress to apply to tissues to obtain an optimal response, the goal of rehabilitation following injury is to enhance the definitive quantity and quality of repair tissue.

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8

Basics of Exercise Physiology

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A good basic knowledge of exercise physiology is necessary when designing and implementing a rehabilitation or conditioning program. The definition of *training*, for the purposes of this chapter, refers to working with the dog in an obedience training program that addresses behavioral aspects, whereas *conditioning* refers to an exercise or workout program that addresses the animal's physiologic capacity to perform work. *Exercise physiology* is a discipline that examines how exercise affects the body and is applicable in the field of physical rehabilitation.

The level of conditioning of a dog constantly changes any time the body performs at a specific metabolic level proportionate to its current adaptation to physical stress (Figure 8-1). The systems of the body are conditioned to maintain homeostasis at this level. If the demands placed on the body increase over time, the body adapts and conditions itself to maintain homeostasis at this new level. An understanding of the physiologic changes involved in this process helps in the development of programs that allow regulation of the conditioning or reconditioning of the body. These programs may be used to condition the entire body or to focus on individual body segments. In physical rehabilitation, programs may be developed to treat injuries, to stimulate healing of the injuries, and to enhance reconditioning of the injured segment. Because of the biomechanical forces acting on the body during the reparative phases of healing, certain segments may become stronger than other segments. Exercise programs should be implemented to remedy any conditioning imbalances of the body. The basic goals should be to allow the tissues to heal, then to recondition the tissues so that they can accept predetermined workloads, and finally to recondition the entire body to balance any conditioning deficits.

Cardiopulmonary Physiology

The circulatory system carries various oxygen, substrates, and chemical messengers; removes carbon dioxide and metabolic byproducts; assists with thermoregulation; and is an important mediator for acid-base balance.¹

Blood consists of fluid and cellular components. Blood plasma is the liquid portion of blood and consists of water, proteins, electrolytes, dissolved gases, and nutrients.¹ The

cellular constituents consist of erythrocytes, leukocytes, and platelets. Oxygen and carbon dioxide (discussed in the next section) dissolve to a limited extent in blood plasma, but are largely carried bound to hemoglobin found in erythrocytes.

The heart serves to pump blood to the pulmonary and peripheral circulation. Resting heart rate is approximately 80 beats per minute for a large dog.¹ In response to exercise, heart rate and myocardial contractility increase, which concomitantly increases cardiac output. A large dog typically has increases of 30% for stroke volume, 300% for heart rate, and approximately 400% for cardiac output during vigorous exercise when compared with rest.¹

Muscle Physiology

Muscle is a tissue greatly affected by exercise or disuse associated with injury. Understanding muscle function and the molecular events of muscle contraction provides a basis for the concepts of physical rehabilitation and exercise physiology. There are three types of muscle in the body: skeletal, smooth, and cardiac. Skeletal muscles connect one bone to another. Each muscle consists of thousands of myocytes. A muscle fiber is long, fusiform, and surrounded by a plasma membrane called the sarcolemma. Inside these fibers are myofibrils, which consist of filaments composed of contractile proteins. These contractile proteins are arranged in units called *sarcomeres*.

Actin and myosin are two types of protein chains in the sarcomere. They interact as a result of enzymatic and chemical reactions to produce muscle contraction. Calcium and phosphate are chemical components involved with the production of muscle contractions. Phosphate is in the form of adenosine triphosphate (ATP). ATP binds to a receptor site at the end of the myosin leverage arm. The actin filament includes troponin, which is bound to strands of tropomyosin. A calcium ion attaches to the troponin molecule, which changes the shape of the tropomyosin. This action opens a myosin-specific binding site on the actin protein chain. Energy is emitted when ATP releases a phosphate ion, producing adenosine diphosphate (ADP). The resultant energy allows the ADP to create a bond between the open binding site on the actin filament and the myosin leverage arm. This bond changes the myosin

structure, allowing leverage to produce a contraction between the two fibers. The ADP is released and the lever arm is freed. Energy is then required to add a phosphate group to the ADP, re-creating ATP, which is then used for further contractions (Figure 8-2). The accumulated contractions of the muscle fibers create contraction of an entire muscle.

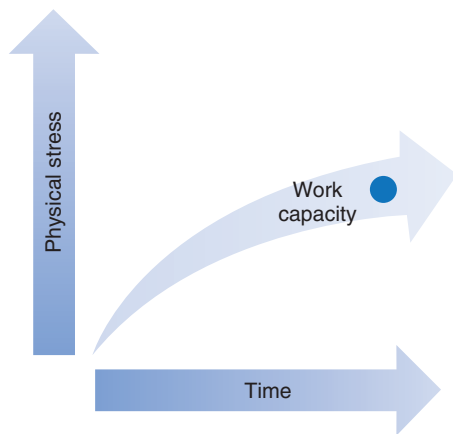


Figure 8-1 Work capacity increases when physical stress increases and adequate time allows for adaptation.

Muscle Structure and Function

The innervation to a muscle controls the muscle contraction by directing an action potential along an efferent motor neuron. A single motor nerve diverges to innervate many muscle fibers. The combination of a nerve and the muscle fiber it innervates is called a *motor unit*. Contraction of an entire muscle is a result of the cumulative contraction of many motor units.

The muscle fibers are grouped together and organized with other fibers by a sheath of connective tissue that is named according to its level of organization. Endomysium covers each of the muscle fibers themselves, and perimysium separates discrete bundles of fibers. Epimysium is the connective tissue layer that surrounds the grouped bundles. These three layers of connective tissue are longitudinally continuous throughout the muscle belly and blend together at each end with the proximal and distal tendons. Muscle fascia is the sheath that covers the epimysium and serves to protect each muscle from movement over hard structures or movement from adjacent muscles. The arrangement of these fibers plays a role in the function of each muscle. It is the combination of motor unit group, the characteristics of the muscle fibers, and fiber arrangement that dictates the resultant type of muscular function (strength, power, or endurance).

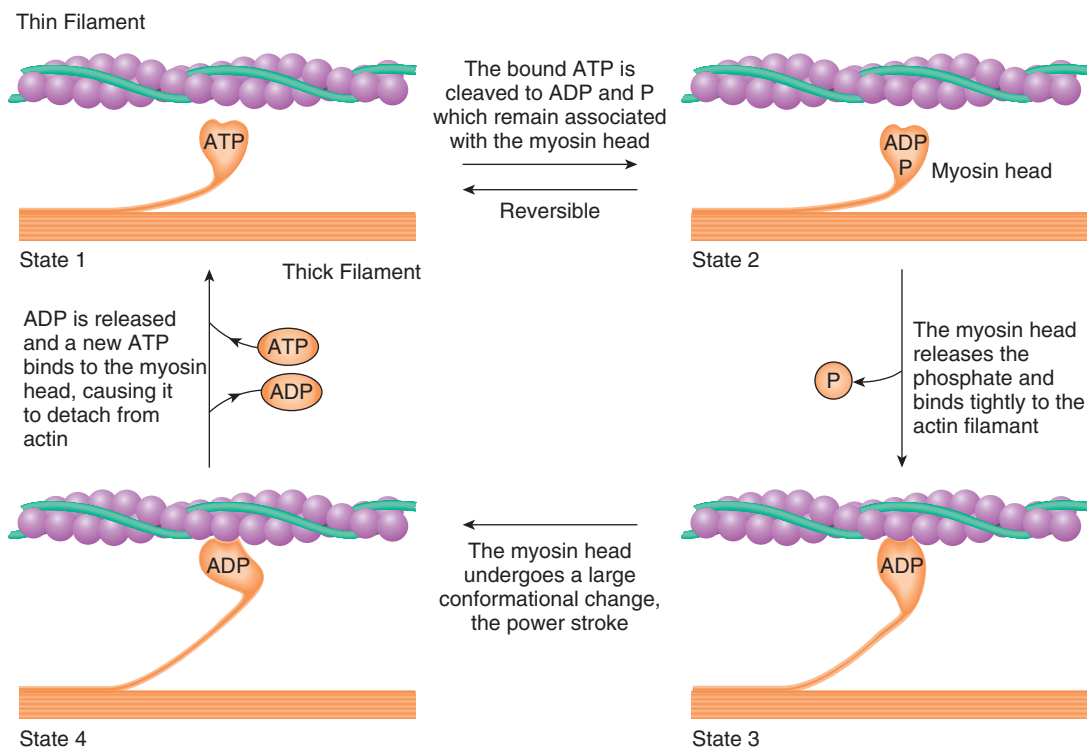


Figure 8-2 Actin and myosin interaction during contraction. ADP, Adenosine diphosphate; ATP, adenosine triphosphate; P, phosphate.

In general, muscle contraction and work are transferred through the tendon and its attachment to a bone. The musculotendinous junction is a layered transition between muscle fibers and the collagen of the tendons. Tendons of origin and insertion may run throughout the length of the structure. Musculotendinous structures are closely tied to functional requirements. Structural shapes of muscle include fusiform and pennate forms. Pennate muscles may be divided into unipennate, bipennate, and multipennate forms. Pennate structures allow a muscle to lift great loads but through a small range of motion, such as in vertebrae. Fusiform structures have the ability to lift a small load at a great velocity through a large range of motion. These types of muscles include the biceps brachii and brachialis muscles in the antebrachium. These two forms of muscle shapes can work together if both strength and speed of movement are needed in a particular joint, such as the shoulder or hip joint.

Muscle Energy Systems

Muscles require energy to maintain basal metabolism and additional energy during physical activity. The body uses three systems to provide this energy: (1) immediate energy sources, (2) glycolytic metabolism, and (3) oxidative metabolism. The type of activity the muscle is performing determines which of the systems will be used. The cellular environment must also be conducive to these processes. Factors affecting the cellular environment include pH, hydration status, temperature, and presence of the proper enzymes. Changes in any of these factors may alter the reactions needed for energy production. While energy production is occurring, metabolic byproducts are simultaneously produced. Byproduct removal is essential, because local accumulation of these substances alters the cellular environment.

The immediate energy source involves intracellular ATP, creatine phosphate (CP), and the ADP-myokinase reaction to provide energy for activity. Intracellular ATP is the first energy source used for contraction. A limited amount of ATP is stored at the myosin crossbridges, near mitochondria, and beneath sarcolemma. ATP breakdown is a hydrolytic reaction and is highly regulated. The body does not allow large changes in ATP content. CP is a high-energy compound used to ensure that the ATP concentration does not become depleted. It is located near the actin-myosin filaments and in the mitochondrial membrane. The CP molecule donates a phosphate group to ADP to replenish the ATP used at the contraction site. The third immediate energy source comes from the myokinase reaction. Myokinase is the enzyme that allows two ADP molecules to combine to form ATP and adenosine monophosphate (AMP). This reaction provides little energy to the system, but AMP serves as one of the allosteric modulators to stimulate carbohydrate (glucose) breakdown in

glycolytic metabolism. Energy from this system lasts anywhere from 5 to 10 seconds with high-intensity exercise and occasionally up to 20 seconds in some elite athletes.

The glycolytic pathway provides energy from 5 to 20 seconds up to 2 minutes as a result of the anaerobic breakdown of glucose. This is a more complex form of energy production, using multiple enzymes and reactions. In the first phase of glycolysis, a glucose molecule enters the cell where the enzyme hexokinase adds a phosphate group to the glucose molecule, which creates glucose 6-phosphate (G6P). G6P then enters a series of reactions to produce fructose 1,6-biphosphate. One of the reactions involves the enzyme phosphofructokinase, which adds another phosphate to the molecule. As a result, two ATP molecules are used in phase 1. In the first reaction of the second phase of the glycolytic pathway, fructose 1,6-biphosphate is converted into two three-carbon molecules. Glyceraldehyde 3-phosphate is then phosphorylated and oxidized, which releases two hydrogen molecules and two electrons. The two electrons and one hydrogen molecule combine with nicotinamide adenine dinucleotide (NAD^+) to form NADH, which can be used in oxidative metabolism. The four remaining reactions of phase 2 result in the production of two ATPs. Combined with two ATPs from the other three-carbon chain, this results in a total of four ATPs produced in phase 2. The net result of the glycolytic pathway is the production of pyruvate and two ATPs that may be used as energy (Figure 8-3).

If the pyruvate is not able to enter the oxidative energy system, it combines with NADH via the enzyme lactate dehydrogenase to produce lactic acid. When released into an environment with physiologic pH, lactic acid releases a proton and becomes lactate. Without a buffer, lactate production results in a decrease in cellular pH. Lactate production in itself is not necessarily detrimental to muscle

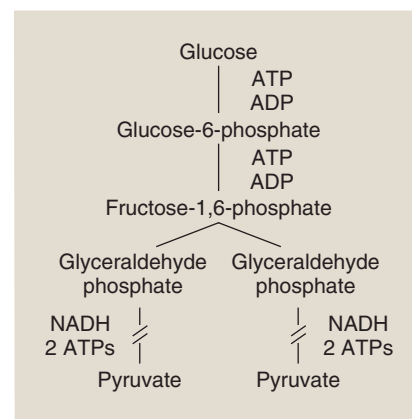


Figure 8-3 Glycolytic pathway with production of pyruvate and adenosine triphosphate (ATP). ADP, Adenosine diphosphate; NADH, nicotinamide adenine dinucleotide, reduced.

metabolism. After bouts of intense exercise, lactate is oxidized back to pyruvate, which can be converted to glucose in the liver, or converted to pyruvate in muscle and other tissues for ATP production. Lactate concentrations coincide with the release of a proton and potential decreases in pH. It is this decrease in cell and blood pH that has detrimental effects on energy metabolism and enzyme activity. Therefore, although associated with the release of protons, the lactate itself is not a problem, but the associated acidosis is.

The third energy source comes from oxidative metabolism, which predominates approximately 2 minutes after the beginning of exercise. It is the most complicated energy system and uses glycolysis, the citric acid cycle, and the electron transport chain. Each of these are complex multiple reaction cascades that result in the production of ATP and energy. Carbohydrates, lipids, and proteins are used as energy sources, and this is the system used to the greatest extent during long-term activity. Pyruvate is converted to acetyl-coenzyme A (CoA) by the enzyme pyruvate dehydrogenase, which also produces carbon dioxide and NADH. Free fatty acids are converted to acetyl-CoA by a process called β -oxidation. The resultant acetyl-CoA then enters the citric acid cycle. The products of this cycle are carbon dioxide, ATP, NADH, and flavin adenine dinucleotide, reduced (FADH).

Oxygen is used in the electron transport chain. The electrons acquired in NADH and FADH are added to hydrogen atoms and oxygen atoms to form water, along with energy to create ATP from ADP. The last electron receiver in the chain is oxygen. This process is termed *oxidative phosphorylation*. Three ATPs are produced for each NADH molecule, and two for each FADH molecule. If oxygen is not present to receive the electrons that flow down the chain, the chain stops, and the result is an accumulation of the components of the electron transport chain and the systems that produce them, halting energy production.

In a rehabilitation program, the conditioning program places energy demands on the muscles. The energy systems used depend on the forces required by the program and the duration of the workout. If the energy systems are insufficient to meet the demands required, the program will not benefit the patient and may actually harm the rehabilitative process. An understanding of muscle physiology will help in designing a conditioning program that is beneficial to the patient.

Musculoskeletal Conditioning

When the body performs at a level greater than its normal daily routine, there is a greater demand for energy. The systems must work together to provide energy to the areas of need, and at the same time maintain homeostasis. Therefore the fuel resources must be sufficient to meet this demand and be available to the body. In other words, energy must be stored in the body to perform physical activity, and the body's energy systems must be conditioned to efficiently

produce, transport, and use this energy in the location where it is required. Homeostatic metabolism is associated with normal daily needs and allows the body to function properly throughout the day. If a short-term stress is placed on the system, the body is able to adapt, within reason, to meet this altered demand. If the stress is repeated on a daily basis, the daily routine is altered and the energy systems are conditioned to meet these new demands. Workout repetition results in adaptation to meet these demands, including pooling energy sources at the location of greatest need (e.g., intracellular ATP, CP, and glucose). The concentrations of enzymes required by the specific energy systems used the most may also increase. Workout repetition conditions the body to the stresses of the new demands and minimizes the chances of systemic or cellular injury.

The muscle response to different types of conditioning programs is determined by the muscle fiber type. The fiber types are divided into type I (slow-twitch) and type II (fast-twitch) muscle fibers. Type I muscle fibers are fatigue resistant, use aerobic metabolism, and have a low contractile force and less ATPase activity. Type II fibers are subdivided by their metabolic characteristics: type IIa (oxidative) and type IIb (glycolytic) fibers. Type IIa muscle fibers use both the oxidative energy system (aerobic metabolism) and the glycolytic energy system (anaerobic metabolism). They may be trained to use one system preferentially according to the demands placed on them. Type IIb fibers primarily use the glycolytic system (anaerobic metabolism), although there is some disagreement over whether dogs have true type IIb muscle fibers as compared with other species.²⁻⁴ These fibers are generally termed type II Dog. Other intermediate fiber types have also been reported. Some change in muscle fiber composition may occur with exercise and conditioning.⁵

How a dog responds to a conditioning program may be affected by its fiber type composition. The type I/type II ratio of the muscle fiber types in a particular muscle is genetically predetermined, although some change in muscle fiber composition may occur with specific conditioning.⁶ Greyhounds bred for sprint activity tend to have a greater percentage of type II fibers than other breeds, whereas foxhounds, which are bred for endurance, have a greater percentage of type I fibers.

In addition to genetic determination, the distribution of fiber types throughout the body and within individual muscles depends on the muscle's function. A higher percentage of type I fibers populate the muscles responsible for maintaining posture, whereas a higher percentage of type II fibers are present in muscles responsible for speed and power.⁷ Altering the muscle function can change the fiber type. Muscle fiber type changes have been produced by tenotomy, tendon transfers,⁸ and altered training.⁹ Knowing the breed of dog and the anatomic area of focus helps determine the type of conditioning program that will be most beneficial for the patient.

There are two general categories of conditioning programs: strength-power programs and endurance programs. In human exercise physiology, endurance events usually last longer than 2 to 4 minutes and are performed at intensities of 90% or less of maximal aerobic power ($\text{VO}_{2\text{max}}$). Strength-power events are of short duration (<2 minutes) and are performed at intensities that are maximal or supra-maximal.¹⁰ The metabolic changes that occur during a conditioning program depend on the principles of intensity, duration, frequency, and method of training. The program should incorporate these principles to result in proper system and organ adaptation.

Central and peripheral physiologic changes occur because of conditioning. Central changes include alterations to the cardiovascular, pulmonary, and endocrine systems. Peripheral changes occur in the musculoskeletal system. The conditioning program chosen should be specific to the type of activity the dog will be performing.

Short-duration, maximal-intensity workouts are recommended if the purpose of the program is to increase strength. This type of program affects peripheral conditioning more than central conditioning. If an increase in speed is the goal, the conditioning program should include sprint workouts at maximal intensity. Workouts that include exercises for the spinal muscles should be included, because it has been theorized that the spinal muscles are the first to fatigue during sprinting.¹¹ Uphill sprint running helps to strengthen and condition the spinal muscles. Running with resistance helps build body strength. Running with a harness and pulling weights is an example of resistance running. In a strength program, the most important change to the muscles is an increase in cross-sectional area because of increased myofibrillar protein content. In addition, there is an increase in muscle resting glycogen and muscle buffering capacity.

If the purpose of the program is to increase endurance, long-duration workouts of submaximal intensity are recommended. Examples are distance running or swimming. An endurance conditioning program may also include some strength activities. Endurance conditioning improves aerobic capacity to provide energy and conditions the musculoskeletal system to use energy more efficiently. In humans, changes occurring in the cardiovascular system include increased stroke volume and cardiac output.¹² Pulmonary changes include an improved ventilation-to-perfusion ratio, tidal volume, and alveolar and arterial partial pressures of oxygen.¹³ Peripheral changes that occur in skeletal muscle are improved capillary density, glycolytic and oxidative enzyme activity, and lactate transport and clearance.¹⁴

Environmental Considerations with Conditioning

Dogs are homeotherms that maintain a normal body temperature between 37.9–39.9°C.¹ Environmental temperatures that exceed body temperature and metabolic heat gain

from physical activity require that the animal employ heat loss mechanisms to maintain homeostasis. Heat transfer mechanisms for gain or loss include conduction, convection, and radiation. Evaporation is solely a heat loss mechanism, and dogs rely almost exclusively on panting for heat loss, especially when ambient temperatures are high.¹ This is because canine sweat glands are poorly developed and largely nonfunctional. Fluid loss from panting can be substantial; thus proper hydration is paramount. Cold environments require the dog to produce more heat via catabolism while conserving heat loss with peripheral vasoconstriction and countercurrent heat exchange.¹

Tissue Injury

Injury of musculoskeletal tissues may occur when a sudden stress occurs that is too great for the body to overcome, or when chronic stresses are greater than the ability of the body to repair itself. This may occur in any of the musculoskeletal tissues, but the type and location of injury depend on the forces and the direction of the forces that cause the injury. For example, a sudden high-energy force exerted on a bone may cause a fracture, whereas chronic prolonged stresses may result in periostitis or stress fractures of the bone. These are examples of bone injuries, but similar stresses may also affect muscle, tendon, ligaments, or synovial tissue.

After injury occurs, the body adapts and the dog alters its movements according to limitations as a result of the injury. If pain is involved, gait is altered to minimize painful movement while allowing locomotion.^{15,16} Injury also changes structure, which alters how forces are received by the body.^{17,18} As the altered movement continues, the musculoskeletal system adapts to produce movement with minimal abnormal stress on the system.¹⁹ However, the altered movement may create additional abnormal stresses that may lead to secondary and sometimes tertiary injuries at locations other than the initial insult.

The altered movement and forces resulting from a musculoskeletal insult should be addressed early before further injury occurs. The tissues involved should be determined by a proper diagnostic workup; then a treatment program that addresses the pain and altered structure is initiated. A physical rehabilitation regimen should be implemented soon after medical treatment is initiated for the initial injury to prevent biomechanical alterations that may result in secondary problems. Secondary injuries complicate the rehabilitative process and may prolong the recovery time.

Further injury may also occur during healing and rehabilitation. If the rehabilitation program places too much stress on the healing tissue, tissue healing may be impaired and the tissue may be reinjured. For this reason, return to normal function should be delayed until the original tissue insult has healed to a point that it can withstand normal activity. A properly managed rehabilitation program should

minimize the time of restricted activity without creating further injury.

Disuse and Immobilization

Following injury and rest, the body adapts to deconditioning, with changes occurring as a result of disuse or immobilization. *Disuse* refers to the state in which the segment is not used as a result of injury or dysfunction. Immobilization occurs when a limb is maintained in a fixed position by splints, casts, or external fixation. Both of these states result in disuse atrophy, which has effects on muscle, bone, ligaments, tendon, and cartilage.^{20,21} Coincidentally, many of the adverse effects associated with aging are a result of progressive reduction of physical activity.

Disuse atrophy of muscle results in decreased circumference and muscle fiber size. Type I muscle fibers are more affected than type II fibers, with a decrease in overall muscle strength. Muscle atrophy and weakness from immobilization occur quickly, and appear to affect anti-gravity muscles to a greater extent initially. Because of the decrease in strength and endurance, there is increased stress on the affected joints, abnormal movement, and an increased risk of injury.

Bone responds with an increase in bone resorption and perhaps decreased bone deposition and is guided by the external forces exerted on the bone. Immobilization also decreases loads on ligaments. Ligament mass decreases after just a few weeks of immobilization so that after 6 to 9 weeks of immobilization, the ligament complexes are only approximately 50% as strong as normal controls.^{22,23} Tendons behave in a similar manner, with alterations in the balance of collagen synthesis and degradation. Articular cartilage also responds by atrophy and thinning of the surface layer of cartilage. If the joint is immobilized in a weight-bearing position, changes similar to those that occur with osteoarthritis may develop.

Immobilization may nevertheless be indicated as part of the initial treatment of many musculoskeletal injuries to allow an adequate healing environment, in spite of the changes that may occur with disuse and immobilization. The detrimental effects of immobilization should be considered in designing the treatment protocol because the recovery pattern is related to the duration of immobilization.

Reconditioning

A reconditioning program should address aspects of tissue healing, musculoskeletal physiology, and joint anatomy. It must include not only the injured area, but also the entire body. Nutrition, psychological aspects, and the effects of tissue injury on the body must also be considered. The goals should be to maximize healing of injured tissue and minimize degradation of healthy tissue. The primary long-term goal of rehabilitation is to maintain the physical condition

of the injured segment during healing and not cause further damage. The secondary goal is to return the tissues of the affected segment to normal strength and character after injury and therapy. The third goal is to recondition the entire body to increase strength, coordination, and endurance to balance any secondary biomechanical disturbances that developed during injury and recovery.

The goal of maintaining condition of the ligaments, tendons, and cartilage may be achieved to some extent by passive range of motion. These tissues maintain their strength or conditioning level by mechanical action or movement, and by the forces placed on them. Bone depends more on loading forces and less on mechanical action and motion. Muscle requires active contractions to preserve tissue strength. In most instances, bone and muscle tissue are most affected during healing and rehabilitation.

After sufficient healing has occurred, the next step is to return strength to the involved tissues. The conditioning program should emphasize strengthening exercises. Before any reconditioning program is begun, the level of strength should be determined, and therapeutic goals should be set by the clinician and owner. The reconditioning program should then be designed to achieve these goals.

There are three types of strengthening exercises: isometric, isokinetic, and isotonic. Isometric exercises involve muscle contraction without movement. This type of exercise is difficult to achieve in dogs. Isokinetic exercises provide constant velocity throughout joint range of motion. Because of the equipment needed to perform this form of exercise, it has not been used in canine rehabilitation. Isotonic exercises involve forces of varying magnitude applied throughout the range of motion and is the most common form of exercise in dogs.

Isotonic exercise involves the muscle contracting while it shortens (concentric) or the muscle contracting while it lengthens (eccentric) while it is loaded throughout a range of motion. Loading the muscle stimulates strength gains. The effort should be high-intensity submaximal loading. Strength gain in reconditioning is similar to that of a regular strength program, except that the initial conditioning status of the tissue is less than normal. Exercises should be focused on strengthening the limb that is most affected by the injury, taking care to avoid overloading the healing tissues to avoid further damage.

As rehabilitation progresses, the focus shifts from localized exercise to overall body conditioning. This includes strength workouts balanced with endurance workouts. The strength portion of the program may still be focused on the areas of weakness, and the endurance portion can act on the entire body.

Nutrition

An important component of any conditioning program is nutrition. Nutritional needs change throughout the

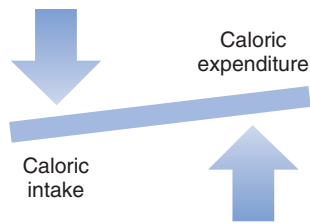


Figure 8-4 Caloric balance depends on intake and expenditure. Body mass is maintained when intake equals expenditure.

treatment and time of rehabilitation. Initially, there is a decrease in energy use by the patient. This should be addressed to keep the patient from becoming overweight during rehabilitation. After reconditioning begins, nutritional components may be added to allow for the nutritional demands of increased physical activity. The basic feeding program should provide for the basic nutritional needs of the patient. The content of protein, carbohydrate, and fat in the feeding program is determined by the type of conditioning. For example, dogs that are performing strength workouts use the immediate and glycolytic energy systems. Therefore they may require a lower percentage of fat in their diets. Caloric balance is the ultimate consideration for diet in dogs (Figure 8-4). If caloric expenditure exceeds caloric intake, then energy will come from various deposits within the body, including stored fat, glycogen, and muscle, and body mass will decrease. Conversely, excessive caloric intake and inadequate energy expenditure is associated with energy storage as glycogen or fat, and increased body mass.

Body mass varies greatly across the variety of dog breeds. Body composition may vary with breed, but largely depends on age. Metabolic rate adjusted for body mass may also vary between some breeds.²⁴ As in human physiology, it appears that canine resting metabolic rate (RMR) declines with age; dogs aged 7 to 8 years have one-half of the RMR of dogs aged 1 to 2 years.²⁴ These factors underscore the importance of a lifetime of physical activity, which stimulates the various physiologic systems and maintains a healthy weight when in balance with caloric intake.

Supplemental feeding should be timed to benefit conditioning. An important feeding strategy is to provide a post-workout snack. This is the optimal time to replenish nutrients that become depleted during the workout. Immediately after the workout, the body is devoted to replenishing the deficits of the energy systems. Feeding a snack during this time promotes replenishment of the deficient nutrients. This snack should consist of simple carbohydrates, a meat protein source, calcium, and phosphorus.

Summary

A proper conditioning program provides a good healing environment for injured tissues. It should minimize the

detrimental effects of injury to the healthy tissues and the entire body. As healing progresses, the program should recondition and strengthen the tissues that are affected by injury. It should also provide nutrients to allow proper healing and be designed to consider the psychological aspects of rehabilitation. An understanding of exercise physiology is necessary to design and implement rehabilitative programs that aid tissue healing and recovery of the patient.

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9

Exercise Physiology of the Canine Athlete

Ralph Millard

Exercise physiology is the study of the effects of exercise and training on normal functions of the body. When implementing a rehabilitative exercise or a conditioning program it is important to have a thorough understanding of normal and abnormal physiologic functions of the body. Sufficient knowledge of physiologic functions of the body increases understanding of the body's response to exercise and allows one to maximize the effectiveness of a rehabilitative program. Exercise has many profound effects on the musculoskeletal system. There is a vast amount of coordination and integration of individual body systems, and, although exercise programs and our rehabilitative efforts mainly focus on improving the function of the musculoskeletal system, it is important to understand the response of other major body systems to the influence of exercise. Physical rehabilitation is typically used to treat or condition areas of the body that are injured or are suffering from abnormal function. Because there may be limitations during healing of an injury, one should understand the body's limitations during injury as well as compensatory methods used by the body. A good general understanding of physiology is very beneficial to the clinician performing rehabilitation, in designing a rehabilitation program and understanding how exercise and training affect the body.

Skeletal Muscle

Muscle Physiology

Skeletal muscle plays a large role in exercise and determining athletic performance, and it is important to understand its structure and function. The muscular system is a collection of contractile units that provide the forces for various functions to take place such as locomotion, posture, respiration, and circulation. The three types of muscle in the body are skeletal, smooth, and cardiac muscle. This section focuses on skeletal muscle physiology.

The functional unit of skeletal muscle is the muscle fiber, or myofiber (Figure 9-1). Most skeletal muscles contain muscle fibers that span the entire length of the muscle, cross over at least one joint, and are typically innervated by only one alpha motor neuron. The sarcolemma is the cell membrane that surrounds each muscle

fiber. Each muscle fiber contains several hundred to several thousand myofibrils. Myofibrils are composed of contractile protein filaments called *myosin* (thick filaments) and *actin* (thin filaments). Myofibrils are divided into segments called *sarcomeres*, each of which lies between two Z discs (filamentous protein different from actin and myosin). From each Z disc actin filaments extend in both directions to interdigitate with myosin filaments.

Contraction of a muscle fiber is first initiated by an action potential, which starts at the central nervous system, travels down an alpha motoneuron to its own axon and the neuromuscular junction. The action potential causes a large influx of calcium ions through calcium channels. The calcium influx results in the release of acetylcholine into the extracellular space between the motoneuron and the neuromuscular junction of the muscle fiber. The acetylcholine then activates nicotinic acetylcholine receptors, opening sodium-potassium channels, triggering an action potential. The action potential travels along a muscle fiber's T-tubule network, depolarizing the muscle membrane. The action potential causes the sarcoplasmic reticulum to release large quantities of calcium into the cytoplasm. A protein called *tropomyosin* is associated with each actin filament. In a resting state tropomyosin covers the active sites on the actin filaments so that attraction between an actin and myosin filament cannot occur and cause contraction. Calcium released from the sarcoplasmic reticulum binds to a protein called *troponin*, which causes the tropomyosin to shift positions and uncover the binding site between actin and the cross bridge portion of myosin filaments. Prior to binding of actin and a myosin cross bridge, adenosine triphosphate (ATP) bound to myosin is cleaved; however, the adenosine diphosphate (ADP) and phosphate ion remain bound to the cross bridge. When the myosin head portion of the cross bridge bonds with the active site on the actin filament, a conformational change is created, causing a power stroke or pulling of the actin filament. The previously cleaved ATP supplies the energy for a power stroke. After the power stroke has been completed, the ADP and phosphate ion are released from the myosin head, allowing for binding of a new ATP molecule. The binding of a new ATP molecule causes detachment of the myosin head from the actin filament. The ATP is then

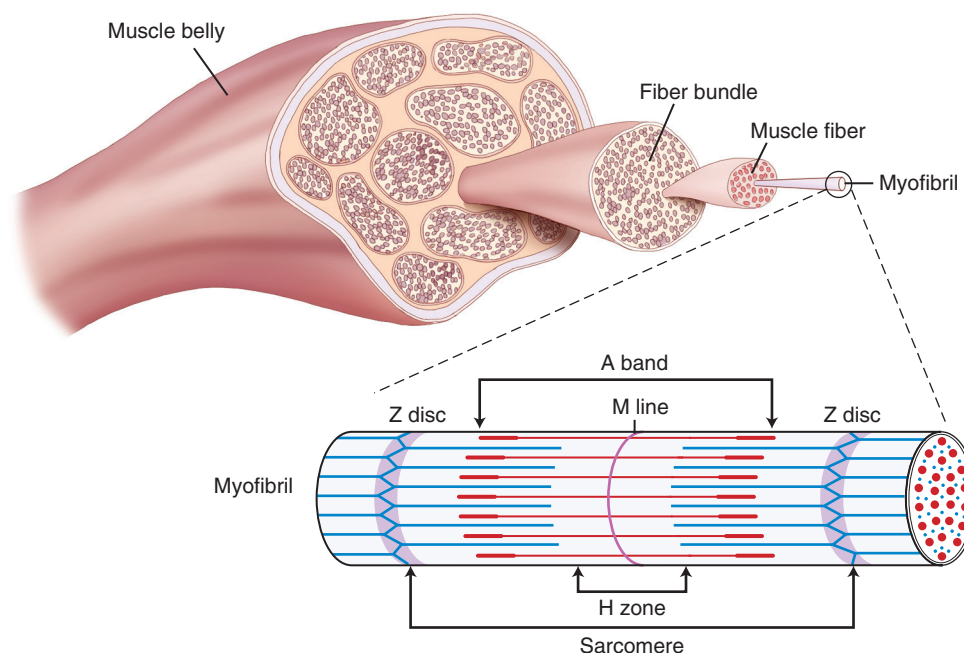


Figure 9-1 Skeletal muscle diagram.

cleaved to begin a new power stroke. This process is repeated again and again to contract a muscle.

Skeletal muscle cells are classified into types based on the specific individual contractile and metabolic properties of each cell. A general classification scheme divides muscle fibers based on their speed of contraction, which is a result of their contractile and metabolic properties. Generally speaking, muscle fibers are considered to be either slow twitch (type I) or fast twitch (type II). Type I fibers contain higher levels of oxidative enzymes, myoglobin, and mitochondria, and are well suited to perform aerobic activities and resist fatigue. Type II fibers have smaller amounts of oxidative enzymes and limited aerobic capabilities. However, type II fibers have large amounts of glycolytic enzymes for anaerobic metabolism and are capable of generating higher levels of tension and shortening velocity, but are much less resistant to fatigue than type I fibers.

Dogs are rather unique in that, along with the aerobic abilities of slow twitch fibers, their fast fibers also have significant oxidative capabilities. In fact, there is significant evidence that suggests an absence of purely glycolytic muscle fibers in the skeletal muscle of the trunk and limbs of dogs.¹ It has been suggested that the only location of type IIB muscle fibers in dogs is in laryngeal muscles.¹⁻³ It is believed that previous studies that relied on immunostaining alone have resulted in misclassification of muscle fibers. Newer studies have implemented electrophoresis, immunoblots, immunohistochemistry, and image analysis to characterize canine muscle fibers.^{1,3}

Currently, five major fiber types are described in the trunk and limb muscles of dogs, which include three pure

fibers (I, IIA, and IIX) and two hybrid fibers (I+IIA and I+IIX).^{1,3} The oxidative/glycolytic ratio of canine muscle fibers decreases from type I (highest) to IIX (lowest). Type IIA is intermediate, with both hybrids between their respective pure isoforms.¹ The decrease in oxidative potential from type I to type IIX is three to four times less than other mammals, which is another finding supporting the superior oxidative capacity of canine skeletal muscle. Even the least oxidative muscle fibers in dogs have a substantial capability for aerobic metabolism compared with fast fibers of other species. The maximum shortening velocity of canine muscle fibers increases from type I (slowest) to type IIX (fastest).³

Muscle Fatigue

Contraction of skeletal muscle at some rates required for exercise cannot continue indefinitely. When a muscle is unable to maintain power output during contraction, it has reached the point of fatigue. There are several explanations for the cause of fatigue. One controversial explanation is that fatigue can originate from the central nervous system (CNS), possibly because of psychological factors that can either enhance or limit performance. Dogs may serve as a good example for this theory because the personality of certain breeds of dogs might allow them to continue to exercise beyond the point that other breeds or species would continue. The drive to run or retrieve might allow some dogs to continue to participate in intense physical activity despite obvious exhaustion, simply as the result of verbal encouragement, whereas other breeds or species would not continue regardless of the stimulus. The

evidence on CNS involvement in fatigue is unclear and unproven.

However, there are several changes at the cellular level of skeletal muscle that result in fatigue during exercise. Fatigue is mainly an imbalance between ATP requirements by muscle and the ability to produce ATP. During exercise, the need for ATP by skeletal muscle dramatically increases. As the components needed to regenerate ATP levels become depleted, there is less ATP available to provide energy for sustained muscle contraction, resulting in an inability to exercise or perform at the same level or rate as before. Phosphate is one of the components needed for formation of ATP. When ATP formation cannot keep up with ATP consumption, excessive inorganic phosphate ions accumulate in the muscle cell. High concentrations of phosphate ions can directly interfere with cross bridging of actin and myosin, as well as inhibit calcium release from the sarcoplasmic reticulum.^{4,5} Lactic acid (lactate) is an end product of glucose metabolism by way of the glycolytic pathway. Lactate is formed during conditions of inadequate oxygen supply to muscle or by muscle fibers with low numbers of mitochondria. Accumulation of lactic acid as a result of anaerobic metabolism within muscle cells can also inhibit enzymes involved in production of ATP.

Glycogen is an important energy source used during exercise. Depletion of muscle glycogen is a critical factor associated with fatigue in elite athletes during prolonged periods of exercise.⁶ Sled dogs are a good example of an elite endurance athlete. High-endurance canine athletes can have a remarkable ability to maintain muscle glycogen concentrations during periods of prolonged exercise. Despite limited carbohydrate intake, sled dogs running 160 km (99.4 miles) per day for 5 consecutive days did not develop cumulative muscle glycogen depletion.^{7,8} On the first day of exercise there was a decrease in muscle glycogen content; however, each of the subsequent periods of exercise resulted in stable muscle glycogen content, which was greater than the content after the first day. It has been determined that muscle glycogen preservation in high-endurance athletes, such as sled dogs, occurs as a result of attenuation of depletion rather than rapid replenishment.⁸ During the first portion of multiday endurance racing, sled dogs use a significant portion of muscle glycogen. On subsequent days they use only a small percentage of their remaining muscle glycogen stores. Rather than glycogen, sled dogs use other energy sources such as muscle triglycerides. Dogs are likely capable of this form of metabolism because of their innate ability to use free fatty acids, their ability to perform high levels of training in endurance activities, and the consumption of a diet with a large percentage of fat as one of its components. Canine skeletal muscle is composed mostly of type I, IIA, and IIX muscle fibers, all of which are capable of high levels of oxidative metabolism.¹ Highly aerobic mammals such as dogs have

an increased capability to use fat-based substrates for energy metabolism.⁹ The high-fat, high-protein diet of sled dogs can help preserve muscle glycogen stores during exercise. Evaluation of fasted rats subjected to several bouts of repeated exercise showed similar results of muscle glycogen preservation, suggesting that metabolic adjustments allow certain species, such as dogs, to preserve muscle glycogen stores during bouts of repeated exercise. There may be a critical concentration of muscle glycogen that the body protects against excessive consumption during repeated exercise. Compared with humans, conditioned canine athletes may have an advantage regarding prolonged exercise based on their highly aerobic metabolism and ability to use alternative substrates for energy and maintenance of muscle glycogen stores.

Conditioning and Training of Skeletal Muscle

Conditioning of skeletal muscle involves performing physical exercise that prepares muscle for the specific task that it will be asked to perform. Training pertains to incorporating the specific exercise into the type of sporting event or activity the dog will be involved in. A portion of training also involves behavior modification required to perform the specific activity. During exercise training athletes are frequently pushed to perform an activity until physically unable to continue. This is commonly termed the *overload principle*, which refers to the need for a system to be exercised to a level beyond that which it is accustomed for training to have an effect. Placing the system in an overloaded state causes it to adapt to new conditions, thereby improving a system's ability to perform a task at a higher level of performance. This principle applies to the cardiovascular system, the musculoskeletal system, and other body systems. Factors that influence overload include intensity, duration, and frequency of a particular exercise. Most types of exercise training can be divided into either endurance or strength training activities. It is important that training uses the systems and structures involved in the activity. This is a training principle called *specificity*. For example, a dog that competes in a sport that requires pulling loads would be better served to run while pulling against resistance as opposed to trotting for a long duration on a treadmill. In humans, engaging in training activities that target specific muscle groups results in adaptations in the muscle types targeted. For example, endurance-type exercises, such as running prolonged distances at a sub-maximal heart rate, increases the capacity to produce energy in an aerobic manner by increasing capillaries and mitochondria, especially in type I muscle fibers.¹⁰

High levels of endurance are very important in dogs that participate in prolonged periods of exercise such as sporting, herding, and long-distance racing dogs. Endurance relates to the ability of a muscle or group of muscles to undergo many repetitions of contraction under low loads.

Endurance exercise usually targets specific muscle groups involved in the specific action and lasts for a prolonged time, usually greater than 15 minutes. Endurance exercises for dogs include trotting, running, swimming, land and underwater treadmill activity, and long-distance sled pulling. It was previously believed that endurance exercise does not result in an increase in type I muscle fibers. More recent research indicates that prolonged endurance exercise can promote a shift of type II muscle fibers to type I in humans.¹¹ Other changes associated with long-term endurance training include an increase in capillary density and vascularity of muscle tissue, increasing the delivery of oxygen and nutrients to the muscle. The issue of muscle fiber type conversion after endurance exercise in humans remains somewhat controversial, but can be explained by the fact that muscle fibers are classified by staining them and as their aerobic capacities are altered, fibers that were borderline in classification can “shift” into another category.

Studies have evaluated skeletal muscle adaptation in dogs during periods of endurance training. One study of foxhounds undergoing endurance exercise failed to show the biochemical and histochemical adaptations in skeletal muscle that have been found in some studies involving humans.¹² Foxhounds did not show alterations in the activity of succinate dehydrogenase (SDH), muscle fiber type, or capillary density of skeletal muscle after a 12-week endurance training program.¹² These findings contrast with those of humans, who do undergo detectable skeletal muscle adaptations after endurance training. SDH is a measure of the oxidative capacity of a muscle. There was no difference in SDH activity, muscle fiber staining, or capillary density in trained and untrained foxhounds, suggesting that an untrained dog is already well suited to support the highly oxidative metabolism of endurance exercise, or that the training program employed in that study was not challenging enough to result in muscle fiber changes. The lack of a change in fiber type staining after endurance exercise may support the theory that dogs, such as foxhounds, are highly reliant on oxidative enzyme metabolism during exercise independent of the pattern of motor unit recruitment.¹²

Another study, however, concluded that endurance exercise can induce a shift of type II to type I muscle fibers.¹³ Beagles exercising on a treadmill 5 days per week for 55 weeks demonstrated an increase in type I fibers in the thoracic spinal muscles as well as the triceps muscle.¹³ The unique nature of canine skeletal muscle as well as the shortcomings of various staining and imaging techniques has made classification of canine skeletal muscle a difficult task, much less classifying changes as a result of endurance training. It is fairly well accepted that canine skeletal muscle has a relatively high capacity for aerobic metabolism.¹ It may be likely that dogs experience changes in muscle fiber type in response to exercise; however, some

of the shared characteristics of type I and type II fibers in dogs have led to misclassification, and therefore uncertainty in potential changes in muscles and their fiber types.¹ Newer techniques used to identify canine muscle fiber types might provide insight into identification of possible fiber type conversion as a result of endurance exercise. In addition to a lack of concrete evidence to support the theory of muscle fiber type conversion, dogs undergoing endurance exercise have not shown an increase in muscle capillary density, capillary to fiber ratio, or capillary area to fiber ratio in some studies.¹² The lack of a change in capillary density in skeletal muscle after endurance training suggests that microcirculation of untrained foxhounds is already adequate to meet the demands of endurance training or the overload needed to promote changes was not achieved in these studies. Even untrained dogs have a high aerobic capacity and appear to be genetically adapted for efficient oxygen extraction and use.

However, it is important not to confuse an untrained dog with a sedentary, inactive, or obese dog. There is a lack of studies involving changes in skeletal muscle in sedentary or overweight dogs after undergoing exercise training. It is likely that untrained but active dogs possess advantages over sedentary or obese dogs at the cellular level of skeletal muscle. Significant biochemical and histochemical changes have yet to be documented in sedentary dogs after undergoing endurance exercise training. However, endurance training is likely very beneficial for the overall health of dogs and improved athletic function. Perhaps future studies may discover previously unknown adaptations of canine skeletal muscle, which are uniquely different from those of humans and other species.

Other adaptations and alterations in body systems occur after endurance training that likely allow for improved performance during exercise. Although there may not be specific biochemical or histochemical change in muscle, there is some support that endurance exercise can improve the functional blood tissue gas exchange properties of skeletal muscle in dogs. A study evaluating endurance-trained dogs and dogs with one limb immobilized found that at a constant rate of oxygen delivery, endurance-trained dogs had an increased $\dot{V}O_2$ max, mainly as a result of increased oxygen diffusive conductance ($\dot{D}O_2$) in skeletal muscle.¹⁴ An interesting discovery, however, was that despite a 31% reduction in muscle weight, there was no difference in $\dot{V}O_2$ or $\dot{D}O_2$ between immobilized dogs and a control group during exercise.¹⁴

Strength of muscle refers to the maximal force a muscle or group of muscles can generate during one repetition. Muscle strength is associated with an increase in muscle size (hypertrophy). Type II muscle fibers are capable of developing more force per cross-sectional area and contract at a higher velocity than type I fibers (Figure 9-2). Strength training increases the size of type II fibers more than type I.¹⁰ High-resistance training increases the number

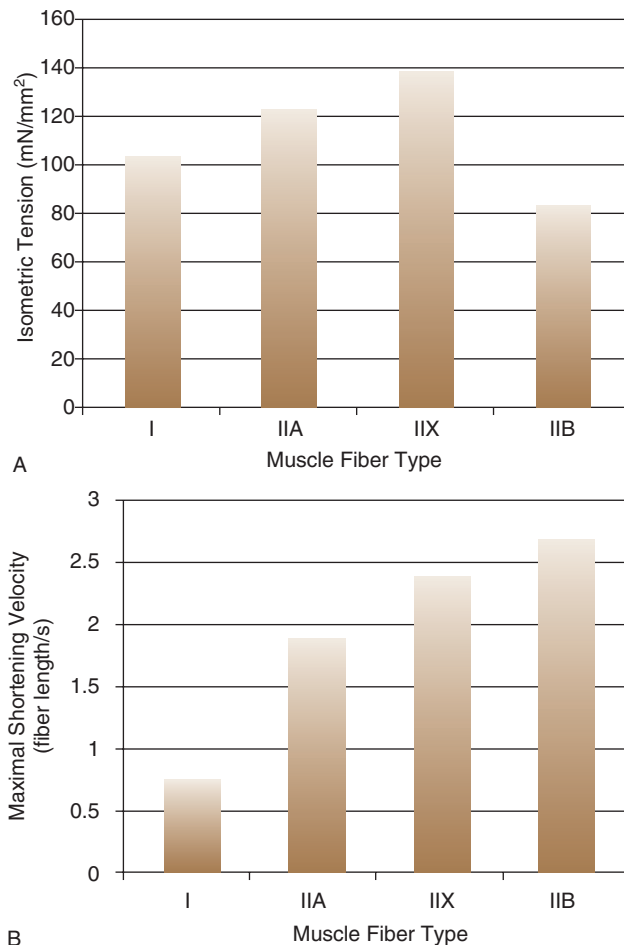


Figure 9-2 A, Comparison of maximal force generated between canine fiber types. B, Comparison of maximal shortening velocities between canine fiber types. (Adapted from Toniolo L, Maccatrozzo L, Patruno M et al: Fiber types in canine muscles: myosin isoform expression and functional characterization. *Am J Physiol Cell Physiol* 292:C1915-C1926, 2007.)

of contractile proteins in type II fibers, which increases the cross-sectional area and the force they are able to generate. The enlargement of muscles that occurs with strength training is thought to mostly occur as a result of hypertrophy of muscle fibers and not hyperplasia (adding muscle fibers). Strength in dogs is mainly related to speed and the ability to carry or pull loads. Strength is also important for dogs that must accelerate and decelerate very rapidly, such as dogs that participate in agility competitions. Like strength, sprinting speed is also most affected by type II muscle fibers. The distribution of certain fiber types also depends on genetic factors. Sprinting breeds, such as greyhounds, have a greater percentage of type II muscle fibers, whereas breeds such as foxhounds have larger percentages of type I fibers and are more adept at endurance exercise. Short-duration, maximal-intensity workout programs are best suited for promoting an increase in strength or speed.

In addition to conditioning muscles of the limbs it is also important to focus on spinal muscles. During sprinting exercise it has been observed that spinal muscles are among the first to experience fatigue.¹⁵ Incline treadmill running and uphill sprinting are good activities to strengthen muscles, including the spinal muscles. Other strengthening exercises include running or racing, pulling or carrying resistance loads, swimming against jets, and trotting or running up inclines or stairs. Performing strength exercise can increase the size of muscle groups and the force they are capable of generating. The opposite is true if skeletal muscle is subjected to periods of disuse or immobilization. Under these conditions, the muscles undergo atrophy and decreased muscle strength of the limb. A study that evaluated muscle atrophy caused by immobilization found that immobilization of the hindlimb for a period of 10 weeks reduced the diameter of type II fibers by 50% and type I fibers by 35% in the vastus lateralis muscle and maximum production of tetanic torque by the affected limb decreased by 50%.¹⁶ When removed from the immobilized state, growth hormones increased the size of muscle fibers and increased strength of the previously immobilized limb during the recovery period significantly more than a control group.¹⁶

Respiration

The term *respiration* can relate to pulmonary or cellular respiration. This section discusses pulmonary respiration and the effect of exercise on ventilation. The term *pulmonary respiration* refers to ventilation of air into and out of the lungs and the exchange of gases between inspired or expired air and lung tissue. *Ventilation* refers to the mechanical movement of air into and out of the lungs. Diffusion is the process by which gas molecules move either from air into blood or blood into air. In addition to providing a means of supplying oxygen to the blood and tissues of the body and allowing for release of carbon dioxide, the pulmonary system plays an important role in regulating acid-base balance during rest and during exercise, as well as thermoregulation. Inflow of air to the lungs occurs as a result of intrapulmonary pressure dropping below atmospheric pressure, and expiration occurs when intrapulmonary pressure rises above atmospheric pressure. The most important inspiratory muscle is the diaphragm; however, any muscle capable of increasing the volume of the chest cavity can contribute to inspiration. Because of the elastic nature of pulmonary tissue and the thoracic cavity, expiration at rest is typically a passive event. During exercise, expiration can become partially active because of contraction of muscles of the abdominal wall.

Precise control of ventilation and pulmonary gas exchange during exercise is required to maintain PO_2 , PCO_2 , and acid-base balance. During exercise a change in ventilation is required to keep up with metabolic needs, which

requires an increase in ventilation.^{17,18} Higher demand for oxygen and a need to rid the body of large amounts of carbon dioxide necessitate higher rates of ventilation. The respiratory control center in the medulla oblongata initiates the drive to inspire and expire. To increase ventilation, input to the respiratory center can come from both humoral and neural sources. At the onset of exercise, one theory states that impulses from the motor cortex to control skeletal muscle movement can “spill” over into the respiratory center as they pass through the medulla, causing an increase in ventilation.¹⁹ This is further supported by the fact that during strenuous exercise, arterial PO_2 , PCO_2 , and pH often remain almost the same as during resting. At the initial onset of exercise, ventilation increases prior to any significant changes in blood chemicals. This supports the theory that initial ventilatory changes are associated with signals sent to the respiratory center of the brainstem at the same time that signals are sent to skeletal muscle. Chemoreceptors in respiratory muscles may respond to potassium and H^+ and send afferent information to the respiratory center. Central chemoreceptors detect increases in PCO_2 and decreases in pH. Peripheral chemoreceptors are sensitive to increases in PCO_2 and decreases in PO_2 or pH. All of these receptors act to fine-tune ventilation during exercise. Input from higher brain centers, regulation of PCO_2 , and lowering of pH caused by increased blood lactate levels are all methods thought to increase ventilation during exercise.

The process of breathing at rest is clearly initiated in the CNS after input from other areas of the CNS, as well as peripheral signals. However, the exact causes of breathing in dogs during exercise prompted investigation to determine whether ventilation is an active or passive process. Some have proposed that the passive forces from limb movement in trotting dogs contribute a substantial amount to hyperpnea during exercise.²⁰ The forces believed to affect respiration include the cranial and caudal displacement of the abdominal organs during acceleration or deceleration; concussive forces of limb impact, which change intrathoracic pressure; flexion and extension of the lumbosacral region resulting in compressive forces within the abdomen thus leading to intrathoracic pressure changes; and inertial forces on the chest wall caused by limb contact with the ground.²⁰

A subsequent study demonstrated different results that contradict the proposition that breathing in dogs during exercise is largely a passive event associated with stride. It was concluded that breathing in exercising dogs is a neural-driven event. Diaphragmatic electromyographic activity, esophageal pressure changes, and phasic diaphragmatic shortening and breathing frequency remain strictly synchronous during exercise and are not significantly associated with stride.²¹ Active recruitment of the diaphragm and abdominal muscles is the major contributor to intrathoracic pressure changes and generation of airflow, and

oscillations of abdominal contents associated with locomotion are not critical for airflow in exercising dogs.²¹

The diaphragm is required to perform very high levels of work to provide adequate ventilation during exercise. The diaphragm muscle is highly oxidative, preventing it from fatiguing easily. Historically, the muscles of the respiratory system were not thought to be a limiting factor during exercise. *Respiratory muscle fatigue* is defined as a condition in which there is a loss in the capacity for developing force or velocity of muscle contraction of respiratory muscles, resulting from muscle activity under load, which is reversible by rest.²² Studies have confirmed that respiratory muscle fatigue can occur during high-intensity exercise in humans.²³ It is also believed that during prolonged high-intensity exercise, respiratory muscle fatigue can limit exercise performance in humans. One notable finding in humans was that fatiguing respiratory muscles cause reflex vasoconstriction to skeletal muscle of the limbs, resulting in exercise limitation of muscles, thus limiting some of the oxygen demand.²³ There have been no studies evaluating respiratory muscle fatigue in healthy exercising dogs; however, the condition occurs in cases of cardiogenic and septic shock.^{24,25} It is likely that the highly aerobic nature of the canine diaphragm muscle normally protects it from fatigue.

Healthy dogs are likely to have a low likelihood of developing respiratory muscle fatigue. However, dogs with underlying conditions, especially those involving the respiratory system, are more likely to be susceptible to fatigue. Dysfunction of the respiratory system, such as pneumonia, increases the work load for respiratory muscles. Dogs engaged in strenuous exercise can reach high rates of ventilation. Although respiratory muscles such as the diaphragm may not fatigue easily during exercise, energy is still required to maintain such a high rate of ventilation. One consideration is that the cost of energy to maintain high rates of ventilation can represent a significant amount of total energy expenditure of a dog during exercise and thus represent a potential limiting factor of performance. A study that calculated the dynamic respiratory work of exercising dogs suggested that the higher rates of ventilation and elevated body temperature associated with exercise may induce changes that decrease the cost of breathing.²⁶ Dogs exercising on a treadmill showed a decrease in dynamic respiratory work as a result of decreased airway resistance that resulted in up to an eight-fold savings of energy expenditure of respiratory muscles. When body temperature rises to more than $39^\circ C$ and respiratory rate increases to 120 breaths per minute, there is an apparent decrease in dynamic respiratory work as a result of lower airway resistance.²⁶ It is unknown whether there is a progressive decrease in airway resistance as body temperature rises or if a sudden change in breathing frequency induces the change. It is believed that the most likely site of resistance reduction is at the level of the larynx. During

states of hyperthermia, cats have reduced laryngeal resistance to almost half of normal values.²⁷ The lower airway resistance in exercising dogs decreases the amount of energy required to maintain the rates of ventilation required for strenuous exercise. This mechanism may help resist respiratory muscle fatigue as well as conserve energy for use by skeletal muscles in the trunk and limbs.

Depending on the environment a dog exercises in, it is possible for ambient temperature to have direct effects on the respiratory system of athletes. Repeated exercise in cold environments can result in airway disease. Dogs exercising in cold conditions can develop pulmonary changes similar to a condition known as “ski asthma” in humans. Dogs exercising for prolonged periods in cold environments show significantly more reactivity to aerosol histamine and overall greater airway resistance when compared with dogs not racing in cold environments.²⁸ This type of dysfunction can persist for at least 4 months after rest. It is believed that continuous cold air hyperventilation causes excessive heat and water loss, leading to airway mucosal injury. The specific effects of cold weather changes of the respiratory system on exercise performance have not been fully evaluated. However, any condition that results in inflammatory changes of the airways would likely have some degree of negative effect on performance.

Cardiovascular System

The primary purpose of the cardiovascular system is to deliver oxygen and nutrients to the tissues of the body as well as remove wastes and aid in temperature regulation. To accomplish these tasks, the cardiovascular system works in conjunction with the respiratory system. During exercise, cardiac output must increase in proportion to the increase in metabolic rate. In human athletes there is a linear relationship between percent maximal oxygen uptake ($\text{Vo}_2 \text{ max}$) and cardiac output. Cardiac output is the quantity of blood pumped into the aorta each minute by the heart and is determined by the product of the heart rate and volume of blood ejected per heartbeat (stroke volume). This in turn determines the amount of blood that flows through the circulation. An increased demand for oxygen and nutrients by skeletal muscles during exercise requires an increase in cardiac output as well as redistribution of blood from inactive organs to active skeletal muscle.

The onset of exercise induces changes that increase cardiac output. The initial changes in the cardiovascular system at the onset of exercise are thought to originate from centrally generated motor signals followed by modification and fine-tuning from baroreceptors, chemoreceptors, and mechanoreceptors throughout the body.²⁹ Both inhibition of the parasympathetic and activation of the sympathetic nervous system act to increase heart rate and strength of contraction during exercise. Stroke volume is also increased during exercise. Venous return to the heart

is increased by venoconstriction, which decreases the storage capacity of the venous system and causes blood to move back toward the heart. The mechanical action of contracting skeletal muscle compresses veins to push blood toward the heart. The rhythmic action of breathing also promotes blood flow from the abdomen to the thorax. Increased flow of blood to the heart increases the volume of blood in the ventricles at the end of diastole (preload) available for the heart to pump back into circulation for skeletal muscle demands. The increased preload also acts to increase the strength of muscular contraction by way of the Frank-Starling mechanism.

Other methods of meeting the increased oxygen demands of skeletal muscle during exercise are to increase blood flow to muscle by decreasing blood flow to other organs and vasodilatation of skeletal muscle arterioles. Skeletal muscles in resting humans receive approximately 15-20% of total cardiac output; during maximal exercise this increases to 80-85% of total cardiac output.²⁹ There is some controversy as to the exact mechanism by which skeletal muscle blood flow increases, and blood flow to other organs decreases during exercise. During exercise, vascular resistance in skeletal muscles decreases while vascular resistance in visceral organs increases.³⁰ Initial vascular resistance within skeletal muscle is believed to decrease as the result of withdrawal or attenuation of sympathetic input to skeletal muscle arterioles; however, maintenance of vasodilatation is likely due to intrinsic control caused by local changes in metabolites such as pH, oxygen tension, and nitric oxide.^{29,31} Another possible mechanism that acts to increase blood flow to muscle in dogs is the concept of *sympatholysis*, which was demonstrated by attenuation of vasoconstrictor response during dynamic exercise compared with a resting state.³²

Studies have been performed regarding the effect exercise has on the cardiovascular system of dogs. Similar to humans, changes in the cardiovascular system of dogs are required to meet the metabolic needs of skeletal muscle during exercise. Sled dogs, some of the most elite canine athletes, have provided invaluable information on the cardiovascular system during maximal exercise. The heart rate of sled dogs can dramatically increase at the onset of exercise. Heart rate can increase from a resting rate of 50 beats per minute to 300 beats per minute and remain between 250-300 beats per minute during a 60-minute run.³³ Foxhounds exercising on a treadmill show similar trends of change in heart rate. Heart rate continues to increase as dogs are subjected in increasing levels of submaximal exercise. When foxhounds reach a state of maximal treadmill exercise (approximately 12 km/hr on a 28% grade), heart rates approach 300 beats per minute (Figure 9-3, A). When comparing heart rate of exercise-trained and untrained foxhounds at different levels of submaximal exercise, trained dogs have on average a heart rate 14 beats per minute lower than untrained dogs at each level of

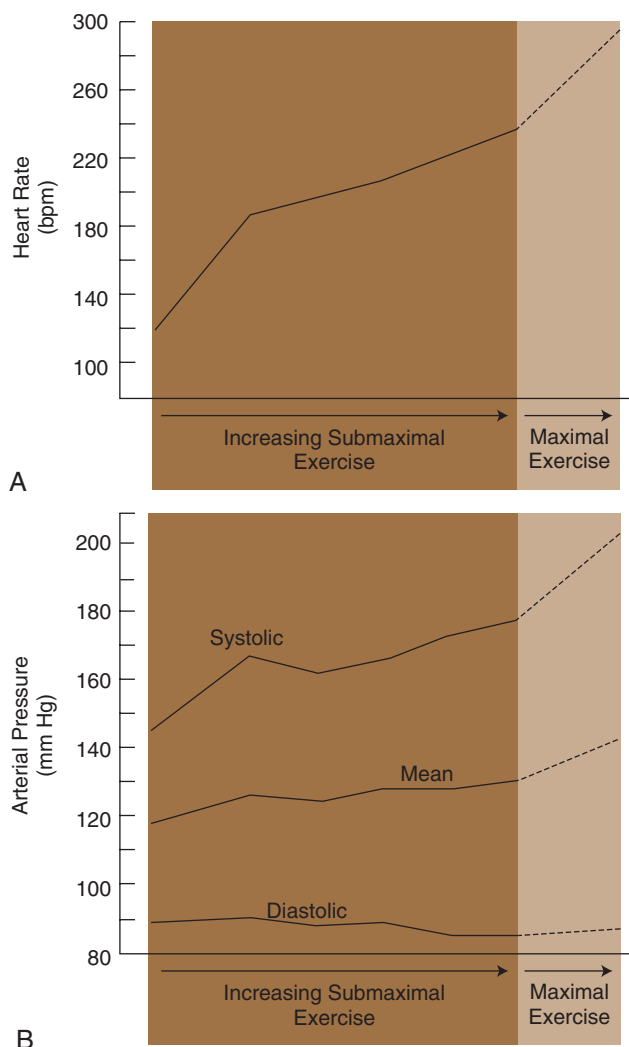


Figure 9-3 A, Changes in heart rate of foxhounds as a function of relative work rate. B, Changes in arterial blood pressure of foxhounds as a function of relative work rate. (Adapted from Musch TI, Haidet GC, Ordway GA et al: Dynamic exercise training in foxhounds. I. Oxygen consumption and hemodynamic responses. *J Appl Physiol* 59:183-189, 1985.)

measured submaximal exercise.³⁴ Exercise also affects blood pressure. Systolic pressures can increase to more than 200 mm Hg during exercise. However, diastolic pressures during exercise are frequently lower than resting values, resulting in a mean pressure similar to that of resting values³³ (Figure 9-3, B). These changes are the result of arteriolar vasodilatation at the level of skeletal muscle, which decreases diastolic blood pressure.

The heart itself undergoes changes as a result of exercise, to better meet demands of the body.³⁵ The term *athletic heart* has been used to describe cardiac changes that occur as a result of prolonged high-intensity exercise. In human athletes, strenuous repetitive exercise

enhances vagal tone and causes left- and right-ventricular enlargement.³⁶ These cardiac changes can result in resting bradycardia, increased end-diastolic volume, and increased stroke volume. Changes such as physiologic murmurs, bradyarrhythmias, and cardiac chamber enlargement are thought to be associated with cardiac adaptations to sustained endurance exercise and not pathologic conditions.³⁶ Auscultatory and electrocardiographic (ECG) studies have found that highly trained dogs have cardiac changes very similar to human elite athletes.^{37,38} Murmurs evident on auscultation are frequently heard in highly trained dogs compared with untrained dogs. Murmurs in athletic dogs are typically grade I-II systolic murmurs heard best over the left heart base.³⁷ The murmurs are likely a result of increased blood velocity passing through the aortic valve in response to physiologic heart hypertrophy and increased stroke volume.³⁷ It has been observed that highly trained dogs without a low-grade murmur tend to perform more poorly than dogs that have a low-grade murmur.³⁷ This is further evidence that low-grade murmurs in endurance-trained dogs are a physiologic response to exercise and not evidence of a pathologic cardiac condition. Cardiac studies on canine athletes have shown that endurance training results in a significant reduction in resting heart rate as well physiologic hypertrophy of the heart.^{38,39} ECG monitoring of canine athletes engaged in endurance training has shown an increase in QRS duration, which is reflective of physiologic cardiac hypertrophy.³⁸ Hypertrophy of the canine heart as a result of endurance training is believed to be a reversible event.³⁸

Blood flow changes in exercising dogs are similar to changes seen in exercising humans. Blood flow to the terminal aorta and coronary vessels of dogs significantly increases during exercise as a result of increased cardiac output.³³ Increased blood flow to the coronary vessels is important because just as skeletal muscle increases its oxygen demands during exercise, so does cardiac muscle. Some studies of splanchnic blood flow in dogs during exercise are in disagreement with one another. One study of blood flow during exercise determined that mesenteric and renal blood flow were unchanged in racing sled dogs compared to resting state.³³ These findings suggested that increased cardiac output and autoregulation of vascular resistance in skeletal muscle, and not redistribution of blood volume from other organs, mainly control increased blood flow to skeletal muscle in exercising sled dogs. This is in contrast with studies in humans that have shown diminished visceral blood flow during exercise as a compensatory change that occurs with redistribution of flow to muscles. A subsequent study involving foxhounds found different results than the sled dog study and discovered that blood flow to abdominal organs decreased during maximal exercise, similar to humans.⁴⁰ The conflicting results may be explained by different blood flow measurement techniques. Blood flow to the splanchnic region of the sled

dogs was evaluated by placing flow transducers around the cranial mesenteric artery and left renal artery. Blood flow of the foxhounds was determined by the use of radioactive labels, which allowed for inclusion of blood flow to abdominal organs from other arteries in addition to the cranial mesenteric and renal. Another variable between the two studies may be effects of ambient temperature on distribution of blood flow. It may be possible that a much cooler ambient temperature influenced splanchnic blood flow to maintain core body temperature in the sled dogs.

Vasoconstriction of inactive tissues and splenic contraction increase oxygen delivery during exercise by increasing blood flow to active skeletal muscle. As previously stated, blood flow to inactive tissues, such as abdominal organs, may be decreased during exercise because of vasoconstriction, and this is mainly a result of alpha-adrenergic influence. Unlike humans, dogs are capable of splenic contraction during exercise as a mechanism of increasing the number of circulating red blood cells and the oxygen-carrying capacity of blood. It is thought that alpha-adrenergic influence also contributes to contraction of the spleen in dogs. One study demonstrated that alpha-adrenergic blockade of dogs undergoing maximal exercise reduced oxygen consumption and increased anaerobic metabolism.⁴¹ The reduced arterial oxygen content was caused by inhibition of splenic contraction of exercising dogs that underwent blockade. The value of splenic contraction in exercising dogs was also demonstrated in splenectomized dogs, which had reduced arterial oxygen content during exercise.⁴¹ Dogs that have undergone splenectomy lack a major adaptation that affects their ability

to undergo aerobic metabolism and influences exercise performance.

Cardiovascular System and VO_2 Max during Exercise

VO_2 max is the greatest rate of oxygen uptake by the body during extreme dynamic exercise and is the best index of the intensity of an aerobic exercise. Maximum oxygen uptake is the product of cardiac output and systemic oxygen extraction. In humans, increases in VO_2 max with endurance training are a result of an equivalent increase in both maximal cardiac output (50%) and systemic oxygen extraction (50%).⁴² Elevation of cardiac output is influenced by increases in preload caused by increased heart size, increased venous return, and increased myocardial contractility, as well as a decrease in the resistance the heart has to pump against (afterload). Cardiac contractility can increase with sympathetic stimulus of the ventricle, as well as by way of the Frank-Starling mechanism. Afterload can be reduced by decreasing peripheral resistance of blood vessels, especially the arterioles supplying skeletal muscle (Figure 9-4). Increased systemic oxygen extraction can result from increased capillary density in skeletal muscle or increased oxidative capacity of muscle cells. Increased blood flow allows for a greater opportunity of oxygen uptake by muscle cells.

Studies of oxygen uptake in dogs have shown some similarities to studies of humans. Foxhounds undergoing submaximal and maximal endurance exercise have a predictable increase in Vo_2 max (Figure 9-5, A). Similar to humans, Vo_2 increases during exercise are attributed to increased cardiac output. The increase in cardiac output in

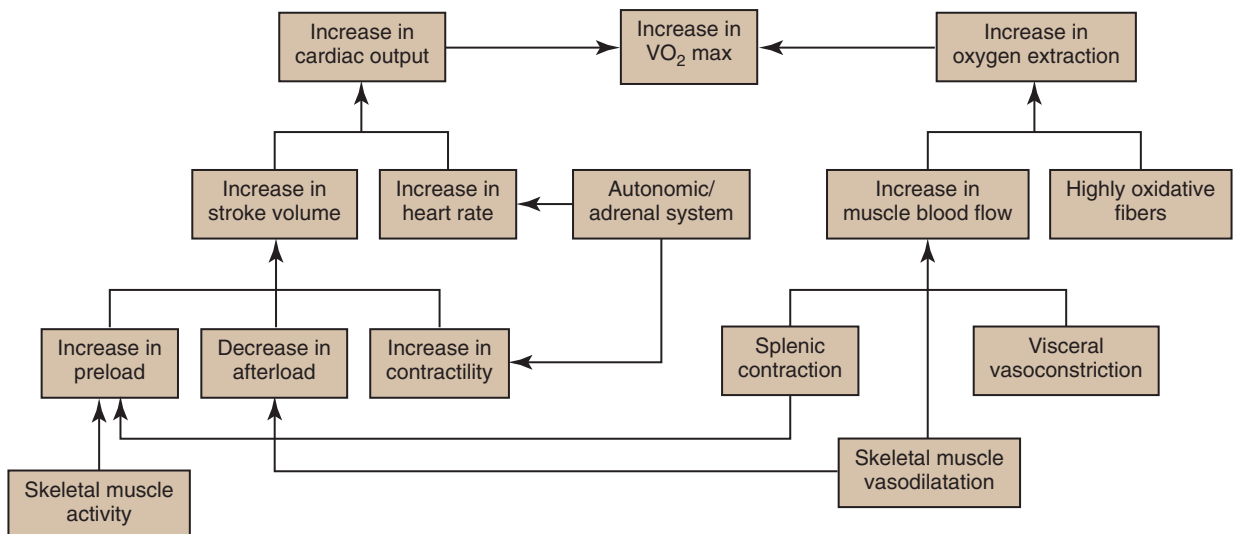


Figure 9-4 Summary of factors affecting VO_2 max during endurance exercise.

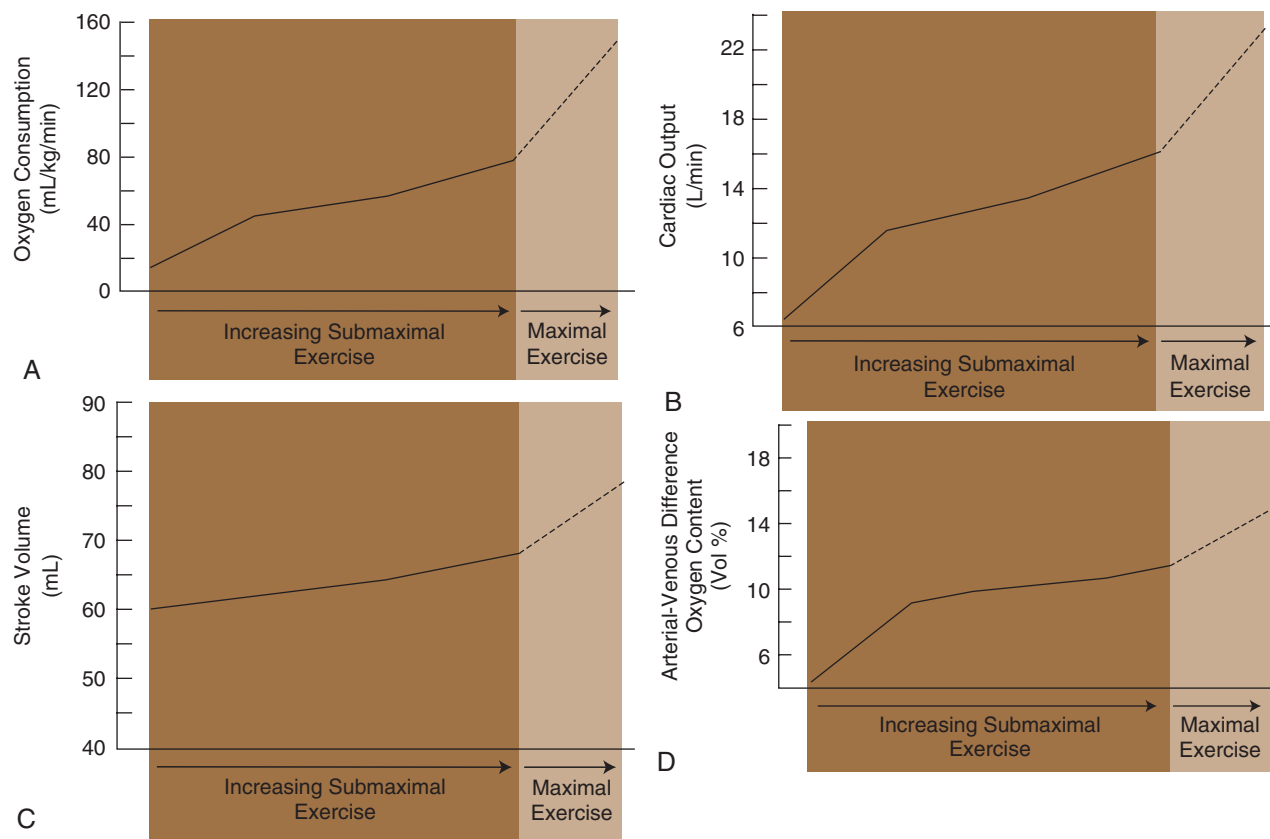


Figure 9-5 A, Changes in oxygen consumption of foxhounds as a function of relative work rate. B, Changes in cardiac output of foxhounds as a function of relative work rate. C, Changes in stroke volume of foxhounds as a function of relative work rate. D, Changes in arteriovenous O_2 difference of foxhounds as a function of relative work rate. (Adapted from Musch TI, Haidet GC, Ordway GA et al: Dynamic exercise training in foxhounds. I. Oxygen consumption and hemodynamic responses. *J Appl Physiol* 59:183-189, 1985.)

dogs during exercise is mostly attributed to increases in stroke volume up to approximately 50-70% of Vo_2 max, after which stroke volume levels off during submaximal activity and further increases in cardiac output are mostly due to increased heart rate³⁴ (Figure 9-5, B and C). In contrast to humans, changes in systemic oxygen extraction do not significantly contribute to increases in Vo_2 max in dogs after undergoing an endurance training program.³⁴ This is further supported by minimal evidence that dogs undergo biochemical or histochemical adaptations in skeletal muscle after undergoing endurance training.¹² Vo_2 max elevation in endurance-trained dogs compared with untrained dogs is derived primarily from central changes (increased cardiac output) and not from changes in oxygen extraction. Increased cardiac output accounts for 96% of the increase in Vo_2 max in endurance-trained foxhounds, whereas increased oxygen extraction accounts for only 4% of the increase.³⁴ Compared with a resting state, dogs engaged in exercise activity have a substantial increase in systemic oxygen extraction (Figure 9-5, D). However, when comparing the oxygen extraction of an untrained and

trained dog during exercise, there does not seem to be significant evidence to demonstrate pronounced increases after training.

When athletes stop training, their Vo_2 max decreases. This is a result of decreased maximal cardiac output and systemic oxygen extraction. In humans, an initial decrease in cardiac output is due to a decrease in stroke volume from loss of plasma volume mainly as a result of a decrease in plasma protein levels and plasma renin and vasopressin activity, which are upregulated during periods of exercise.^{43,44} As a state of detraining continues, Vo_2 max continues to decrease, along with reduced systemic oxygen extraction. In humans, periods of detraining decrease the overall oxidative capacity of skeletal muscles.⁴⁵ Because untrained dogs have an oxidative capacity similar to dogs that have undergone endurance training, it is reasonable to speculate that a decrease in Vo_2 max of a previously trained dog that has entered into a period of detraining would mostly be a result of a decreased cardiac output and not the reduced ability of skeletal muscle to extract oxygen.

Exercise Capacity and Age

The age of a dog can affect the ability to exercise. Older dogs have reduced capability for maximal exercise capacity.^{46,47} A group of older dogs (10-14 years) showed significantly reduced exercise capacity and maximum oxygen consumption compared with a group of younger dogs (2-3 years) during treadmill exercise.⁴⁶ The reduction in Vo_2 in older dogs was mostly associated with reduced cardiac output.⁴⁶ Decreased Vo_2 in older exercising beagles was attributed to a 25% decrease in cardiac output and an 11% reduction in systemic oxygen extraction.⁴⁶ The relative reduction in cardiac output in older dogs appears to be caused in part by attenuation of heart rate during exercise as well as stroke volume. These findings correlate with a study that demonstrated a reduced maximal heart rate response to isoproterenol in older dogs.⁴⁸ A reduction in maximal heart rate of older adult humans during exercise has been demonstrated in numerous studies.^{49,50} The reduction in stroke volume in older beagles may be a result of age-related changes to the myocardium, peripheral vasculature, pericardium, or decreased splenic contraction during exercise.⁴⁶

In addition to cardiovascular changes in older dogs, changes in skeletal muscle can have an effect on exercise performance. Aging is associated with a decrease in muscle mass and capillary density. Older beagles have been noted to experience a 16% loss of type II muscle fibers when compared with younger dogs. The total fiber area of type I muscle fibers seems to be maintained, and may even increase in some areas, such as in the semitendinosus muscle.⁴⁷ There is also a reduction in muscle capillary density and capillary/fiber ratio in older dogs. Despite these age-related changes at the level of skeletal muscle, blood flow is relatively well maintained during exercise. During maximal exercise in older dogs there is no significant decrease in blood flow to locomotion muscles.⁴⁷ A significant decrease in vascular resistance to skeletal muscle and reduction of blood flow to splanchnic circulation seem to allow for maintenance of muscle metabolism during exercise. Despite maintaining blood flow to skeletal muscle during exercise, older dogs have a reduced maximal exercise capacity compared with younger dogs. In summary, the maximal Vo_2 for an older dog is significantly less than that of a younger dog at maximal exercise, but reduction in blood flow to muscle does not appear to be a factor for decreased Vo_2 max in older dogs. Rather, the main causes are reduced cardiac output and oxygen extraction.^{46,47} Maintenance of blood flow to skeletal muscle during exercise only partially compensates for age-related changes such as decreased metabolic potential, oxygen content of arterial blood, and cardiac function, which account for the inability of older dogs to reach the same level of Vo_2 max as younger dogs.⁴⁷ It should be expected that the exercise performance of dogs will likely decrease

with advancing age. Exercise performance may also be more likely to decline in older dogs with the onset of other medical conditions that may develop with advancing age.

Exercise and Nutrition

Nutrition requirements for athletes are different from those who lead a more sedentary lifestyle, and the nutrition requirements of athletic dogs are different from that of human athletes. Compared with humans, canine skeletal muscle contains a high degree of oxidative fibers. The majority of exercise studies on dogs have been performed on either greyhounds or sled dogs. However, information from effects of diet on the performance of athletes, such as sled dogs, has been applied to diet modification of less specialized athletes. One common ingredient in the diet of canine athletes is raw meat. Raw meat by itself is not a balanced diet. It is deficient in essential vitamins and minerals and if fed alone can lead to conditions such as nutritional secondary hyperparathyroidism and increased risk of fracture.⁵¹ Calcium deficiency is the most common imbalance in a raw-meat diet. Addition of 1 g of bone meal to every 50 g of raw meat can correct the deficiency; however, excess supplementation can increase the risk of orthopedic disease, especially in larger immature dogs.⁵² Risk of food-borne infection is another concern when feeding a raw-meat diet. Common pathogens such as *Salmonella*, *Escherichia coli*, *Campylobacter*, *Listeria*, and Shiga-like toxin produced by *E. coli* O:157:H7 are potential pathogens found in raw meat that can pose a very serious hazard to both dogs and humans.⁵³ Therefore, raw-meat diets are not recommended for canine athletes.

Daily energy requirement for maintenance depends on body weight, age, exercise, breed, and ambient temperature. Cooler temperatures require less energy for maintenance. However, higher levels of energy are required once the ambient temperature drops below the point at which metabolic rate must increase to maintain body temperature. For thin-coated breeds this temperature is approximately 20°C and for thick-coated breeds critical temperature is less than 0°C.⁵⁴ Energy requirements for movement are more proportional to the distance traveled than the speed of travel.⁵⁵ It requires more energy for a dog to trot 13 miles than it does to sprint 100 yards at top speed. Certain diets may increase stamina during exercise. Exercise stamina for humans can be increased by high-carbohydrate diets, which increase muscle glycogen levels.⁵⁵ Dogs are different than humans in that they metabolize free fatty acids at twice the rate of humans.⁵⁶ Skeletal muscles in dogs are more adapted to use fat than humans. A high-fat, low-carbohydrate diet increases the stamina of an exercising dog compared with a diet with only a moderate amount of fat.⁵⁷ High-carbohydrate diets increase muscle glycogen levels; however, during short high-intensity exercise, glycogen is used up more rapidly in dogs. Fat oxidation during

both rest and exercise is twice as active in dogs as compared with other less aerobic species.⁹

High levels of protein are another important component of the canine athlete. Anemia can develop in highly athletic dogs fed a low-protein diet. A diet of at least 32% of energy as protein prevents a drop in hematocrit in highly athletic dogs.⁵⁸ For sprint exercises, mild restriction of food intake before races to reduce body weight improved sprint performance in greyhounds.⁵⁹ Different than endurance racing dogs, feeding an increased protein and decreased carbohydrate diet to sprinting greyhounds resulted in slower performance times. However, when higher levels of fat, along with high protein and lower carbohydrate levels, were fed to greyhounds, race times improved.^{60,61} To increase the stamina of working or exercising dogs, canned diets with high levels of fat and protein can be added to a commercial diet before periods of heavy exercise. An alternative is to add beef to a commercial diet that has high levels of energy as both protein and fat.⁵⁵ It may be necessary to supplement a dog's diet with components such as fat and protein to meet the energy needs during exercise; however, it is very important that the diet be balanced and complete for all needs of the body.

The components of an athletic dog's diet are important, but the timing of feedings is important as well. Strenuous exercise in untrained dogs delays gastric emptying for more than 1 hour.⁶² Feeding of meals should not occur during periods of exercise, and it is best to wait until after a short period of rest.

There is little information on the vitamin, mineral, and other nutrient requirements for athletic dogs. Commercial dog foods contain sufficient amounts of vitamins and minerals for nonathletic dogs. If excessive supplements are added to a commercial dog food, toxicity issues are a possibility.

Temperature

Regulation of body temperature is a critical process to maintain homeostasis and normal function. Metabolic pathways and cellular structures are affected by temperature. Prolonged or excessive increases in body temperature can denature proteins needed for vital metabolic pathways and can even result in death. A decrease in body temperature can slow metabolic pathways to the point that vital body functions are affected. Body temperature is determined by the difference between heat gain and heat loss. Maintenance of body temperature is a highly regulated process. The thermoregulatory center is located in the anterior hypothalamus. It receives information from peripheral thermoreceptors to initiate changes in the body to either increase heat production or to increase loss of body heat. Normal metabolic processes in the body produce heat. At rest, the metabolic rate is low and the production of heat is relatively small. During intense exercise, heat production

can be quite large. The body is relatively inefficient at using energy, and a significant portion of energy use for work results in heat production.

During periods of intense exercise it becomes necessary for the body to take measures to increase heat loss to prevent body temperature from rising too high. Heat loss can occur by four different mechanisms: radiation, conduction, convection, and evaporation. *Radiation* is the loss of heat in the form of infrared rays. This process involves the transfer of heat down a thermal gradient from the surface of one object to another with no physical contact. *Conduction* is the transfer of heat from a warmer object to a cooler object that it is in contact with. *Convection* is the loss of heat into either air or water that is in contact with the body. *Evaporation* is the loss of heat as it is transferred to water on a surface of the body and converted into gas. By far the most important method of heat loss for exercising dogs is evaporation. Dogs have poorly developed sweat glands, which are not significant in thermoregulation. Instead, dogs mostly use evaporation as a method of heat loss by way of panting. During exercise there is increased blood flow to the respiratory system, tongue, and oral mucosa, and an increase in saliva and respiratory secretions. Blood has a high capacity to store heat and is very efficient at transporting heat. Exercising dogs take frequent shallow breaths, allowing air to rapidly move over the oral mucosa and respiratory dead-space for evaporation and heat loss.

Effect of Ambient Temperature on Exercise

Despite countermeasures taken by the body, dogs can experience significant increases in body temperature during periods of exercise. Healthy Labrador retrievers performing field trial retrieving exercises for 10-minute periods outdoors can have a marked increase in body temperature.⁶³⁻⁶⁵ In an ambient temperature of up to 28°C, normal healthy Labrador retrievers undergoing strenuous exercise can have temperature elevations up to 42°C.^{63,66} An increase of up to approximately 2°C more than resting temperature is common during normal exercise among Labrador retrievers under these types of circumstances. In addition to hyperthermia, dogs typically experience a respiratory alkalosis, hypocapnia, and mild metabolic acidosis when exercising in temperatures ranging from 11-29°C.^{63,64} Another study confirmed that Labradors retrievers undergo a significant increase in temperature during strenuous exercise regardless of a cool or warm ambient temperature.⁶⁵ However, despite no severe differences in body temperature in strenuously exercising dogs under the previously mentioned range of ambient temperatures, deleterious changes may occur when exercising in warmer environments.

Dogs exercising in an ambient temperature less than 21°C were less likely to develop respiratory alkalosis and hypocapnia. Dogs exercising in an ambient temperature greater than 21°C are very likely to develop respiratory

alkalosis and hypocapnia.⁶⁵ In an experimental setting, severe acute hypocapnia results in decreased cerebral and myocardial perfusion. It is uncertain what effect these hemodynamic changes have on healthy dogs during exercise. Several dogs in these studies became uncoordinated and weak during the training exercises; however, they recovered and did not experience any further problems. Sprinting greyhounds show similar dramatic increases in temperature, but in contrast to Labrador retrievers, they tend to have a more pronounced drop in blood pH after racing.⁶⁷ The lower blood pH in greyhounds may also be partially explained by the fact that greyhounds experience significantly higher blood lactate elevations during exercise.

It seems that in ambient temperatures between 21–29°C, despite significant increases in body temperature and hemodynamic changes, healthy dogs do not experience significant adverse clinical effects. Although there were no serious adverse clinical effects to the dogs performing retrieving trials in these studies, care should always be taken when exercising dogs, especially in warmer conditions. Sporting breeds such as Labrador retrievers often have a personality (drive to retrieve), which causes them to voluntarily exercise to the point of exhaustion. It is possible that uncontrolled excessive exercise in an environment with a high ambient temperature can lead to adverse effects such as heatstroke. Dogs with underlying conditions (especially respiratory related) would likely be much more susceptible to adverse effects of exercising in warm environments.

Exercising dogs should have access to plenty of water to help avoid possible dehydration as well as an area for the dog to retreat to for a cool down if necessary. If a dog shows signs of significant fatigue, it is advisable to allow for a period of rest before resuming exercise. A study that compared dogs that were cooled with ice packs during exercise with dogs that were not cooled showed that the cooled group had increased exercise performance, and lower heart rates, respiration rates, and blood lactate levels than the noncooled group.⁶⁸ Owners and trainers should monitor dogs closely for signs of heatstroke during exercise and seek immediate veterinary attention if necessary.

When exercising for a short period in a cool ambient temperature, changes in cardiac output and blood flow mainly address increasing blood flow to working skeletal muscles. When exercise is sustained long enough to raise body temperature, or when exercising in a higher ambient temperature, blood flow must change to allow for heat loss in addition to supplying the demand of skeletal muscle. Peripheral pooling of blood potentially leading to reduced cardiac output is a cost of cooling for humans exercising in the heat.⁶⁹ Humans performing heavy walking exercise in the heat can experience a significantly reduced cardiac output.⁷⁰ Perhaps yet another physiologic adaptation of dogs is that exercise in warm environments results in a

higher degree of cardiac output.⁶⁶ The cooling mechanism of dogs does not involve peripheral evaporation from the skin. Because of this, dogs do not experience peripheral pooling of blood during exercise in heat, and thus do not experience the drop in cardiac output that humans can experience. Dogs seem to experience much larger rates of heat storage compared with humans. It is common for body temperature to reach 42°C in exercising dogs, which does not typically result in adverse effects in healthy animals. This type of exercise hyperthermia is also thought to be an effective method of water conservation during exercise in warm weather. This well-tolerated elevation in body temperature in dogs reduces the amount of water lost to evaporative cooling, which would occur at a higher rate if body temperature were maintained at a lower level during exercise.^{71,72} Dogs that experience prolonged periods of extreme inactivity may be more susceptible to development of severe hyperthermia compared with dogs that are active on a routine basis.⁷³

Dehydrated dogs that exercise have significantly higher body temperature and lower rates of water lost by evaporation. Cardiac output and carotid blood flow are significantly reduced as a result of decreased plasma volume.⁷¹ When dehydrated, the process of reducing evaporation is a method to save total body water. However, the body temperature of dehydrated animals could become dangerously high. Dogs in a state of dehydration should be withheld from exercise, and ample water should be supplied for exercising dogs to help prevent a state of dehydration.

Adverse Effects of Exercise

Exercise and training are designed to condition the body, enhance the performance of physical activity, and improve the overall state of fitness and health. However, the act of exercise can potentially have adverse effects on the body. The body can certainly become injured as a result of external influences, such as the force of impact on the body, or variables in the environment. However, the act of physical activity alone can unmask an underlying disease or condition or it can directly affect an otherwise healthy body by causing alterations that have the potential to have adverse effects.

Unexpected injury and even death, although unlikely, can happen to dogs that are elite athletes as well as those that lead a more sedentary lifestyle. There is limited information on causes of severe injury or death of dogs during exercise. One study's aim was to evaluate necropsy findings on Alaskan sled dogs that died unexpectedly during competition. The most common postmortem findings included aspiration pneumonia, gastric mucosal lesions, and blood loss as a result of gastrointestinal ulceration.⁷⁴ Other findings included rhabdomyolysis, hepatocellular necrosis, and cardiac myodegeneration. The cause

of pneumonia in most cases was aspiration of gastric contents. All dogs with pneumonia had concurrent gastric lesions and it is thought that the lesions found in the gastrointestinal tract play a part in causing vomiting or gastroesophageal reflux, leading to aspiration. Severe injury or sudden death is a rare event and does not occur frequently during periods of exercise.

Serious injury or death during exercise is more likely to be related to an underlying condition that was exacerbated by the demands and stress associated with exercise. It is possible that signs of an underlying condition could go unnoticed during periods of inactivity. Certain conditions can manifest signs for the first time during periods of exercise. During periods of strenuous exercise, Labrador retrievers can be affected by a syndrome called exercise-induced collapse (EIC). This condition is a fairly common cause of exercise intolerance in otherwise healthy young adult Labrador retrievers. Dogs affected by this condition can tolerate moderate amounts of exercise, but may become ataxic and collapse after only 5-15 minutes of strenuous exercise. The cause of EIC has been determined to be a mutation in the dynamin 1 gene.⁷⁵ This gene plays a role in synaptic vesicle endocytosis of neurotransmission in the brain and spinal cord. The first signs of this condition are gait abnormalities, such as a wide-based pelvic limb stance along with rocking and loss of balance. If a dog is known to have this condition, exercise should be discontinued immediately if gait abnormalities are recognized. All dogs in one study had loss of patellar reflex after exercise.⁷⁶ Other conditions that may first be noticed during exercise and can lead to very serious problems or even death include malignant hyperthermia, hypoglycemia, laryngeal paralysis, myasthenia gravis, cardiac abnormalities, metabolic disturbances, and others.

The gastrointestinal tract may experience abnormal changes during exercise in otherwise healthy athletes. Gastrointestinal disease, such as diarrhea, is a common abnormality in human endurance athletes, including marathon runners. Sustained periods of exercise increase the frequency of gastric erosions and ulcerations.⁷⁷ Studies have also shown that in addition to gastric lesions, prolonged strenuous exercise can cause intestinal mucosal damage, increasing intestinal permeability.⁷⁸ It seems that gastric mucosal damage can occur at more intense levels of exercise (submaximal and maximal); however, intestinal permeability can also be altered during less intense routine training exercises.⁷⁹ It is not known specifically what effect, if any, these lesions have on the performance capability of elite athletes such as sled dogs. Administration of famotidine once daily decreases the severity and perhaps the prevalence of exercise-induced gastric ulcers in dogs.⁸⁰ Although the effect of gastric ulcers on performance is unknown, the ease of administration and lack of adverse effects may make the use of famotidine as prophylaxis against gastric ulcers beneficial in performing dogs.

Hematologic and Biochemical Effects from Exercise

Dogs undergo various hematologic and biochemical changes as a response to exercise. Most changes are not severe, and it is often not clear if the changes are truly abnormal or if they are normal physiologic responses to exercise.

Prolonged endurance exercise results in a catabolic state that can cause changes in hydration status, electrolyte alterations, plasma enzyme values, and other hematologic changes. Sprinting and agility dogs can experience changes in blood lactate levels, pH, hematocrit, protein, and potassium levels. However, the changes are not severe and do not appear to have a significant effect on exercise performance.^{81,82} Well-conditioned sled dogs develop multiple abnormalities in serum chemistry values, but they are of limited severity.⁸³ These changes are most likely attributed to adaptation to prolonged endurance training and effects of dietary components such as high amounts of fat and protein. One of the most significant changes in racing sled dogs is the development of hypoproteinemia, mostly caused by hypoglobulinemia.⁸³ This change may be caused by immunosuppression, elevated protein catabolism, and protein loss in the gastrointestinal system as a result of exercise. Short-duration strenuous activities result in activation of platelets and neutrophils in dogs.⁸⁴ Activation of neutrophils is a process that occurs to prime the cells to fight against infections. Activated neutrophils are beneficial in septic conditions. In a nonseptic state, such as during exercise, activated neutrophils may have adverse effects. Activated neutrophils upregulate surface receptors and are primed for release of oxygen radicals and proteolytic enzymes, making them less likely to migrate to extravascular sites and perhaps more likely to lodge in small vessels in body.⁸⁵ Neutrophil activation has been documented in humans as well, but there are no data to support adverse effects of neutrophil activation on exercise performance. A concern with activated platelets is development of a hypercoagulable state; however, this state has not been demonstrated in exercising dogs. It is possible that increased fibrinolytic activity balances out platelet activation during exercise. Exercise training can induce a decrease in packed cell volume. This may be explained by plasma volume expansion and blood loss secondary to gastrointestinal tract bleeding.⁸⁶

Chronic exercise training leads to expansion of plasma volume.⁸⁷ This may be an adaptive mechanism to provide a larger vascular volume to meet the increased cardiovascular and thermoregulatory needs during exercise. Greyhounds can increase their plasma volume by almost 30% after undergoing a 2-week training period.⁸⁸ Studies suggest that the primary reason for plasma volume expansion in endurance dogs is due to an increase in water intake with minimal involvement from renal control.⁸⁸

Agility dogs can increase their hematocrit by 21%, plasma volume by 4%, and total blood volume by 12%

during exercise as a result of splenic contraction.⁸⁹ Within minutes of discontinuing exercise activity, the hematocrit decreases as a result of recapture by the spleen. Agility exercise in dogs does not seem to induce changes in red blood cell indices or morphologic findings.⁸²

Exercise can also affect hormones such as thyroid hormones. Short-term treadmill exercise and sprinting increase levels of thyroid hormones.^{90,91} Prolonged endurance exercise results in substantial decreases in plasma hormone levels (T4 and T3).⁹² The effect of changes in thyroid hormone on exercise performance in dogs is unknown. It is important that the finding of abnormal thyroid hormones in an exercising dog does not lead to a potential misdiagnosis such as hypothyroidism.

Lactate concentration in blood increases during periods of exercise. The degree of increase is related to the intensity of activity as well as the index of glycolytic activity in skeletal muscle. Different breeds of dogs partaking in various forms of physical activity experience differences in the degree of lactic acid elevation. Lactate concentration in dogs performing in an agility competition rise to approximately 4.5 mmol/L, which is almost twice the resting value. Fifteen minutes after discontinuing the exercise, lactate levels are not significantly different than resting values.⁸² Foxhounds engaged in maximal exercise and Labrador retrievers participating in field trials develop lower concentrations of lactate during exercise at approximately 2.7 mmol/L and 2.5 mmol/L, respectively.^{34,64} The difference in lactate levels may be due to different levels of aerobic metabolism between foxhounds and Labrador retrievers, and the dogs participating in the agility competitions, or that the agility competitions are much shorter in duration and are a more anaerobic activity. A more dramatic contrast is seen when comparing other breeds of athletic dogs with greyhounds, which frequently experience a rise in lactate levels close to 30 mmol/L.⁸¹ Short periods of maximal exercise experienced during racing certainly contribute a significant portion of this increase; however, the higher percentage of glycolytic muscle fibers in greyhounds may also explain this breed difference. The blood lactate elevations do not appear to have a negative effect on healthy dogs participating in these types of exercise activities.^{34,64,81,82} Exercise can also result in alterations of various serum chemistry values in dogs. Successive days of endurance exercise can cause increases in serum sodium, creatine kinase, aspartate aminotransferase, alkaline phosphatase, and alanine aminotransferase, and a decrease in serum potassium.⁸³ In healthy dogs these changes are mild and are not of significant clinical importance.⁸³

Exercise and Mental Well-Being

Exercise is an important aspect of life for both humans and animals. The association between exercise and physical

health is widely accepted in the human medical world.⁹³ There is growing evidence that exercise has benefits regarding the mental well-being of humans as well.⁹⁴ In humans there is evidence that exercise can improve physical self-perceptions, enhance mood states, help treat clinical depression and anxiety, and even improve cognitive function.⁹⁴ In addition to the physical health benefits such as weight loss in obese dogs, and increases in strength and endurance, exercise may also have some psychological benefits in dogs.

Collecting information on the mental well-being of dogs is not easily done, leaving perceptions of owners as the only means of determining their well-being. The term *quality of life* is frequently used when attempting to describe the well-being of a dog. The owner, or the individual who spends the most time with a dog, is frequently the one who makes assessments regarding quality of life. Many people would agree that it is easy to see the delight a dog experiences when participating in various forms of exercise, whether it be retrieving, agility, sled racing, and so on. A quality of life screening program of owners evaluated aspects of a dog's life that were thought to be major influences of quality of life. The most common suggestion from owners was that more exercise for their dogs would improve their quality of life.⁹⁵ Exercise is not only a beneficial method of improving physical health and implementing therapeutic effects for rehabilitative purposes, it is an experience that often provides great enjoyment for dogs.

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SECTION III

Patient Assessment and Management

10 Orthopedic and Neurologic Evaluation

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Detection and assessment of orthopedic and neurologic disorders may be challenging. Ideally, each person involved in the treatment and rehabilitation of the patient should be able to perform a physical examination for evaluation of orthopedic and neurologic conditions, and should have knowledge of the underlying condition and its management. The evaluator should develop a systematic approach to standardize the physical examination and to prevent omissions of important findings.

History

The patient's age, gender, and breed should be recorded in the medical record. It is helpful to obtain information regarding the patient's general health. The owner should be questioned to determine if there are any signs of systemic disease, such as fever, anorexia, depression, vomiting, or diarrhea. The presence of any preexisting conditions that may produce or contribute to the lameness should be recorded.

Any known previous trauma should be recorded, along with the affected areas that were injured. The travel history of the affected animal or of other pets in the household should be assessed. Certain fungal diseases and some diseases transmitted by ticks, such as ehrlichiosis and Rocky Mountain spotted fever, are endemic in certain geographic areas, and travel to those regions may result in infection and lameness.

The owner or handler should be asked to state the chief complaint, or the reason why the dog is being evaluated. When questioning an owner or handler regarding the affected limbs, it is important to be certain that the correct limb is identified. Sometimes, the owner may be confused and misidentify the affected legs. To eliminate any confusion, the owner should point to the limb that is the source of concern.

The progression of the condition should be addressed, including when it started, changes in severity over time, changes with weather, changes with exercise or rest, and any changes that occur over the course of a typical day. Information regarding any previous diagnoses should also be obtained. The results of any previous treatments should be assessed. In particular, any medications administered should be recorded, along with the dose, dosing interval, length of treatment time, and whether or not the treatment was effective. It is relatively common for inappropriate doses of medications, such as nonsteroidal antiinflammatory drugs, to be administered, and it may be inappropriate to conclude that treatment has been ineffective in the treatment of conditions such as osteoarthritis.

The owner's impression of the degree of pain or lameness should be obtained, and whether there is multiple limb involvement. With regard to the lameness, the following short series of questions may be useful:

- How long has the lameness been present?
- Has the lameness increased or decreased in severity?
- Is the lameness worse in the morning or evening?
- Was the limb subjected to some traumatic event?
- Does the lameness improve or worsen after a few minutes of activity, such as walking?
- Does the lameness change with weather or exercise?
- Have other limbs been involved?
- Have there been any previous related diagnoses or treatment?

During the gathering of historic information it is vital that the examiner also consider that certain neurologic or oncologic problems may mimic orthopedic diseases, and the converse is true as well. Dogs with bilateral cranial cruciate ligament ruptures may have gait and stance problems suggestive of neurologic disease. Oncologic conditions, such as osteosarcoma or multiple myeloma, may present with a lameness that mimics an orthopedic problem.

Some neoplastic conditions may have paraneoplastic manifestations, such as hypertrophic osteopathy, in which a mass in another location may result in periosteal reaction of the bones of the distal limbs and lameness.

General Physical Examination

It is critical that a thorough general physical examination be performed by the attending veterinarian. Information concerning other body systems is vital to determine whether certain forms of therapy may be contraindicated, or if a systemic disease such as Lyme disease or systemic lupus erythematosus may be the underlying cause of a lameness. It is especially important if surgical intervention is possible or if long-term medication is anticipated to be certain that major body systems are healthy and able to tolerate the treatment. In particular, the cardiovascular, respiratory, gastrointestinal, hepatic, renal, and endocrine systems should be assessed to determine if underlying disease conditions are present that may affect the selection and appropriate use of various medications or anesthetic agents. Thorough auscultation of the heart and lungs, abdominal palpation, lymph node palpation, evaluation of mucocutaneous junctions, and assessment of the skin are necessary. Ancillary tests may be needed for more thorough evaluation, including electrocardiogram, complete blood count, serum biochemistry profile, urinalysis, radiographs or sonograms of the thorax or abdomen, and other special tests.

Initial Observation

The dog should be carefully observed at rest and at several gaits before in-depth examination and palpation to avoid artificially accentuating lameness with manipulation of the affected areas. If possible, observe the patient as it arises from a sitting or recumbent position. Often the lameness is more severe immediately after rising and then improves with ambulation; in some instances, the patient may not use the affected limb at all to help rise to a standing position.

With the patient standing, one should observe for weakness, limb trembling, asymmetry of regions of the limbs indicative of muscle atrophy, asymmetry of the head and neck, limb position, and conformation ([Figure 10-1](#)). It is common for standing animals to bear less weight on a limb afflicted with lameness. In this situation, the animal may not have the entire foot in contact with the floor. One method to assess the relative amount of weight placed on each limb is for the examiner to place the front feet or the rear feet on the palm of each hand and allow the animal to bear weight on the observer's hands rather than on the floor. This is a semiquantitative method of assessing weight bearing. Another method to assess weight bearing is to use bathroom scales, or more sophisticated devices described in Chapter 13, under each foot to determine the amount of weight placed on each limb at a stance. In many lamenesses there is also marked muscle atrophy of the affected



Figure 10-1 The dog should be observed from (A) the front and (B) the rear while standing for evidence of reduced weight bearing on an affected limb, turning the toes out, and not bearing weight symmetrically in comparison with the contralateral limb. Note the reduced weight bearing on the left rear limb indicated by less contact of the metatarsal pad with the ground and the more lateral position of the foot as compared with the other rear limb.



Figure 10-2 The dog should be observed going (A) away from and (B) toward the evaluator at both a walk and a trot. (C) The dog should also be observed from the side.

Box 10-1 Lameness Score (Applied Separately at a Walk and a Trot)

- 0:** Normal
- 1:** Slight, intermittent lameness
- 2:** Obvious weight-bearing lameness
- 3:** Severe weight-bearing lameness
- 4:** Intermittent non-weight-bearing lameness
- 5:** Continuous non-weight-bearing lameness

leg. In some chronic lamenesses there may be abnormal foot pad wear and unequal weight bearing that may be noticeable.

Gait

Gait evaluation is generally performed after observing the dog rising and at a stance. The patient's gait is best observed in a large, enclosed exercise area from a distance. In addition, it is beneficial to have an area in which observations may be made with the dog off leash. Gaiting begins with the handler moving the patient first at a walk, then at a trot. The patient should be evaluated while moving directly away from, and then toward the examiner (Figure 10-2). The patient should then be evaluated from the side at a distance (Figure 10-2C). Evaluation at a lope or gallop may be performed if the patient is able, but this gait is less useful for evaluation of most lameness because of the speed of limb motion and the fact that these gaits are asymmetric. It may be helpful to capture the gait with a video camera and then evaluate the gait in slow motion. The degree of lameness may be subjectively evaluated using a lameness scoring system to allow comparison of the lameness over time (Box 10-1).

Observation of the patient while it is walking is beneficial for those patients with severe lameness that may be unwilling to trot. Because the walk is a slow four-beat gait, each limb may be separately evaluated. This is particularly beneficial to help differentiate a lameness seen at a trot, in

which a diagonal forelimb and hindlimb strike the ground at the same time. Observing at a walk with separate limb strikes helps to determine which diagonal limb is affected in cases of subtle lameness. In many cases, the head will nod up and down if a lameness is present. In general the head will nod up when the affected leg is placed on the ground if a forelimb lameness is present, and the head will nod down when the affected limb is placed on the ground if a hindlimb lameness is present. The reason for this is to reduce weight bearing and force on the lame limb by using the head and neck as a lever arm to shift weight from the painful limb to the opposite end of the body.

The patient should also be evaluated at a trot, which is a symmetric two-beat diagonal gait. Subtle lameness is generally easier to see at a trot because greater forces placed on the lame limb accentuate the lameness. Although moderate forelimb and hindlimb lameness may be differentiated at a walk because the pattern of footfalls is separated, differentiation of forelimb and hindlimb lameness is more challenging at a trot. The reason for this is that diagonal forelimbs and hindlimbs strike the ground at the same time. As an example, consider a head that nods up when the left forelimb strikes the ground. This may indicate a left forelimb lameness. However, it may also be indicative of a left hindlimb lameness, because as the left hindlimb strikes the ground, the head will nod down; as the right hindlimb and left forelimb strike the ground immediately after, the head will nod back up. Careful observation of other features of gait may help distinguish between forelimb and hindlimb lameness at a trot, in conjunction with other features of the physical examination. It may also be helpful to place each major joint through a full range of motion, and then reassess the lameness. Stresses placed on joints as they are flexed and extended may accentuate a subtle lameness.

In addition, other factors such as stride length, limb carriage, joint motion, and side bending of the spinal column are useful in evaluating lameness. The affected limb may have a shortened stride length and reduced

flexion and extension of the affected joints. Most dogs will walk with the normal limb centered under the body during the weight-bearing phase of gait on that limb, whereas a lame limb may be carried further eccentrically during weight bearing. The affected limb may also circumduct during gait. Animals with hip or stifle conditions may have increased lateral flexion of the spinal column toward the affected side in an attempt to advance the limb by using the back, especially if the stride is shortened on that side. In all cases, symmetry of movement should be evaluated. Abnormal proprioception, dragging of the nails, knuckling over on the dorsum of the paw, and hypermetria indicate neurologic disease.

When possible, the patient should be walked and trotted in a large circle in both directions. In many cases, lameness of the inside limb will be accentuated. Stairs and steps may also be helpful in observing lameness and subtle neurologic problems. Dogs may skip up or down steps with the affected limb rather than pushing off the limb to ascend or descend a flight of stairs, or may knuckle over on the dorsum of the paw of a limb afflicted with a neurologic condition.

General Palpation for Symmetry and Atrophy

Following observation of the patient for lameness during ambulation, a brief examination is performed while the dog is still standing to assess symmetry and to assess relative muscle mass and the presence of muscle atrophy. Simultaneous palpation of both forelimbs and then both rear limbs will allow the examiner to detect subtle differences between limbs that are normally symmetric (Figure 10-3). Fractures and neoplasia of the musculoskeletal system are usually obvious. Individual regions are evaluated for swelling, abnormal shape, heat, and sensitivity and pain.

Beginning with the forelimb, the dorsal border of the scapula is palpated to be certain that one side is not higher than the other. The spine of the scapula is assessed for fractures and prominence of the spine of the scapula, which might indicate muscle atrophy if it is readily palpable and differs from the other side. The acromion and greater tubercle of the humerus are palpated next. The distance between the two landmarks and the amount of soft tissue between the two points should be equal on both sides. The shaft of the humerus is palpated next to locate any areas of swelling or muscle atrophy.

The elbow is a difficult joint to assess because many structures are present in a relatively small area, including the medial and lateral epicondyles of the humerus and the flexor and extensor muscles that arise from them, the head of the radius, the region of the medial coronoid process of the ulna, the joint capsule, and other soft-tissue structures in the area. The task is made easier by standing



Figure 10-3 Simultaneous palpation of both hindlimbs for evidence of muscle atrophy and asymmetry.

behind the dog and placing the thumbs of each hand on the lateral epicondyles. The fingers are then wrapped around the cranial aspect of the elbow, and the index fingers are placed in the region of the medial coronoid processes of the ulnas. The area of the medial coronoid process is firmly pressed to assess for pain and discomfort, which may indicate a fragmented medial coronoid process. Following palpation of this area, the cranial and caudal aspects of the medial joint are assessed for joint effusion, swelling, and increased or decreased soft-tissue mass. The caudolateral joint compartment is assessed next between the lateral epicondyle and the olecranon process for joint effusion or swelling, which might indicate osteoarthritis or an ununited anconeal process. This swelling is usually palpable caudal and parallel to the lateral epicondyle, and compression forces the effusion medially. By palpating the elbows simultaneously with the dog standing, subtle abnormalities may be detected by assessing the area for asymmetry. It may also be helpful in certain cases to pronate and supinate the distal extremity when evaluating each individual elbow. These maneuvers help to concentrate the forces in the medial or lateral joint compartment, accentuating any subtle discomfort.

The shafts of the radius and ulna are palpated from proximal to distal to assess the bones for any pain or swelling. The carpi and digits are usually best assessed with the dog in lateral recumbency, although the standing angle of the carpus should be observed to be certain that carpal hyperextension is not present.

The hindlimbs are evaluated next. The dorsal aspect of the wings of the iliums are located to be certain that one is not located further cranially or dorsally than the other, which might indicate a sacroiliac luxation. The gluteal muscles are assessed for atrophy throughout the ilium. The area between the greater trochanter and the tuber ischium is located and compared with the other side. Asymmetry may indicate the presence of muscle atrophy on the side that has less soft tissue between the two points, or a hip luxation if the distance between the two points differs. The thigh musculature is evaluated for atrophy. The stifles are simultaneously palpated in a manner similar to that of the elbows to assess for joint effusion, the presence of firm tissue on the medial aspect of the distal femur (often referred to as a *medial buttress*) that might indicate a chronic rupture of the cranial cruciate ligament, and the position of the patellae, which should be symmetric.

The tibias are assessed for any asymmetric swelling or pain, and the angle of the tarsus is observed to evaluate the integrity of the common calcaneal tendons. The tarsus and digits are best evaluated with the patient in lateral recumbency.

Orthopedic Examination

A systematic approach to the orthopedic examination is necessary to reduce the risk of missing a condition and to create an evaluation that can be consistently repeated. In many cases the contralateral limb may be used as a reference for comparison of limb circumference, range of motion, and sensitivity to palpation. Comparison with the contralateral limb may be especially helpful when a subtle condition exists and there is uncertainty regarding an abnormality. Concentrating on anatomy during the examination, and referring to an anatomy book as necessary, helps in confusing or unusual situations. The patient is usually examined in lateral recumbency, but may be evaluated while standing if it is anxious.

Rear Limbs

Evaluation of the rear limbs first may be safer for the examiner while the patient is becoming accustomed to the examiner. The phalanges and metatarsals of the distal limb are evaluated first. The toes are spread apart to evaluate the nail beds, webbing between the toes, and the pads for any infection, trauma, or cracks. The phalanges, interphalangeal joints, metatarsal bones, and metatarsophalangeal joints are individually palpated and assessed for pain, crepitus, swelling, fractures, luxations, or collateral ligament instability (Figure 10-4). The range of motion of the joints is also assessed. The area over the plantar sesamoid bones is palpated for pain and proliferative changes that might be associated with an old fracture or arthritis. Joint stability



Figure 10-4 The interdigital areas, pads, phalanges, and metatarsal bones should be palpated and the joints placed through a range of motion.

is further assessed by flexing and extending individual joints and placing varus and valgus stresses on the joints. It is helpful to be familiar with the origin and insertion of the interphalangeal ligaments to properly evaluate instability (Box 10-2).

The tarsus is a complex series of joints. It is first assessed for joint effusion and instability. Palpation of the dorsal aspect of the tarsocrural joint is especially rewarding in cases of joint effusion. The joint is placed through a full range of motion to assess the joint for limitations, crepitus, and pain. The examiner's finger is used to palpate the tarsal bones, with particular attention to the central, third, and fourth tarsal bones. The tarsus has short and long components of the medial and lateral collateral ligaments that are taut in different degrees of flexion and extension. To assess the long components of the collateral ligaments, valgus and varus stresses are applied while the tarsus is placed in full extension (Figure 10-5). Normally, there should be little varus or valgus movement with the hock extended. The short portions of the collateral ligaments are assessed by placing varus and valgus stresses on the tarsus with the tarsocrural joint flexed to 90 degrees. Assessment of these structures while the hock is flexed is more difficult because there is some normal motion in this position. After stabilizing the metatarsus with one hand and the tuber calcis with the other, dorsal and plantar stresses are placed on the hock to determine if subluxation is present or if there has been damage to the supporting plantar structures. With the stifle maintained in an extended position, the hock is flexed to evaluate the calcaneus and common calcaneal tendon. Normally, there should be little ability to flex the tarsus. Excessive flexion may indicate that there has been a fracture of the calcaneus or damage to the common calcaneal tendon. It is also helpful to

Box 10-2 Common Conditions of the Hindlimbs

Common Conditions of the Digits, Pads, and Feet

Paronychia
Lacerations of the interdigital skin or foot pads
Luxations of the interphalangeal joints
Luxations of the tarsometatarsal joint
Fractures of the metatarsal bones
Fractures of the phalanges
Fractures of the sesamoid bones

Common Conditions of the Tarsus

Osteochondritis dissecans of the talus
Fracture of the calcaneus
Fracture of the central tarsal bone
Fracture of the third or fourth tarsal bones
Luxations of the tarsus with damage to the short or long tarsocrural ligaments
Osteoarthritis
Disruption of the common calcaneal tendon
Rupture of the plantar ligaments

Common Conditions of the Tibia and Fibula

Fractures of the tibia and fibula
Panosteitis
Traumatic periostitis
Osteosarcoma

Common Conditions of the Stifle

Cranial cruciate ligament disruption
Osteochondritis dissecans of the femoral condyle
Injuries to the medial and lateral collateral ligaments
Medial and lateral patellar luxation
Medial and lateral meniscal damage
Fractures
Congenital absence of the patella
Caudal cruciate ligament disruption
Osteoarthritis

Common Conditions of the Femur

Fracture of the femur
Panosteitis
Osteosarcoma

Common Conditions of the Coxofemoral Joint

Hip dysplasia
Fracture of the acetabulum
Fracture of the femoral head or neck
Fracture and separation of the proximal capital femoral physis
Legg-Calvé-Perthes disease
Traumatic luxation of the coxofemoral joint
Osteoarthritis

palpate the insertion of the tendon on the calcaneus to rule out partial avulsion injuries, which may be manifested as a firm swollen area. It is valuable to compare the suspected limb with the contralateral limb to assess subtle findings.



Figure 10-5 The tarsus is evaluated for evidence of medial or lateral collateral ligament damage by applying varus and valgus stresses to the joint.



Figure 10-6 The tibia is palpated for evidence of pain, swelling, or inflammation.

Palpation of the tibia is relatively easy because the medial aspect has little muscle covering. The metaphyseal and diaphyseal regions are palpated for periosteal or bone pain, which might indicate hypertrophic osteodystrophy, panosteitis, fractures, neoplasia, or traumatic periostitis (**Figure 10-6**). It is helpful to palpate the medial and lateral malleoli distally and the head of the fibula proximally.

The stifle joint is frequently afflicted with orthopedic conditions. Sedation may be necessary in some larger patients that are tense. The joint is initially assessed for swelling. The tibial tubercle is located first, and the path of the patellar ligament is traced proximally. The joint is palpated medial and lateral to the patellar ligament. Normally, the ligament feels like a pencil and the medial and lateral edges can be distinctly felt, but with joint effusion, the patellar ligament is less distinct and fluid may be palpable in cases of moderate to severe joint effusion. The femoral condyles and the region of the trochlear ridges are palpated next. In some cases, osteophytes or joint capsule thickening may be palpable. Thickening of the medial

aspect of the distal femur is often present in dogs with chronic rupture of the cranial cruciate ligament. This is easily felt by placing the examiner's hand and fingers across the stifle so that the fingers rest over the medial surface of the joint. The other stifle may be simultaneously palpated to compare the two limbs. Swelling of the cranio-lateral joint in a young dog might indicate the presence of an avulsion of the long digital extensor tendon, although this condition is rare. The stifle is flexed and extended to evaluate for crepitus, grating, clicking, or snapping, which might indicate osteoarthritis, a damaged meniscus, or joint instability. The examiner's hand may be placed on the patella to assess crepitus. Often dogs with a partial rupture of the cranial cruciate ligament or chronic osteoarthritis secondary to cranial cruciate ligament rupture have marked pain with full stifle extension.

The patella is assessed for medial or lateral luxation. To test for medial patella luxation, the stifle is extended, the distal limb is internally rotated, and pressure is applied to the lateral aspect of the patella to try to displace it medially (Figure 10-7). While maintaining pressure, the limb is slowly flexed and extended to see if luxation occurs with the limb in a position other than full extension. Medial patella luxation is common in toy and miniature breeds and in larger breeds such as Labrador retrievers. In small dogs, the patella may be difficult to locate if patella ectopia exists. In these cases, the tibial tubercle is located and the path of the patellar ligament is traced proximally until the



Figure 10-7 To evaluate the stifle for medial patella luxation, the stifle is extended, the tibia is internally rotated, and an attempt is made to push the patella medially out of the trochlear groove.

patella is located. To test for lateral patella luxation, the stifle is placed in extension or slight flexion while the distal limb is externally rotated. Pressure is applied to the medial aspect of the patella, trying to force it laterally. Although medial patella luxation is most common in all breeds, if a lateral patella luxation is present, it is most likely a large dog or a chondrodystrophic breed.

Instability of the medial collateral ligament is assessed with the stifle in extension. While holding the distal femur with one hand and the proximal tibia with the other, a valgus stress is placed on the tibia. In normal dogs, little motion occurs with this maneuver. Lateral collateral ligament integrity is assessed by holding the limb as if checking for medial collateral instability, but a varus stress is placed on the tibia.

Injury to the cranial cruciate ligament is common, and this structure should be carefully evaluated. Large dogs may be tense and require sedation for a thorough examination. The most common method of assessing rupture of the cranial cruciate ligament rupture is the cranial drawer test (Figure 10-8). To perform this test, place one index finger on the tibial tubercle and stretch the soft tissues caudally with the thumb until the head of the fibula is reached. Place the other index finger on the patella and stretch the soft tissues with the thumb until the lateral fabella is palpated. Now place the stifle in mild flexion and quickly and gently try to move the tibia cranially and caudally (slide the drawer open and closed). The examiner should avoid internally rotating the limb while performing the drawer maneuver. Because the cranial cruciate ligament also limits excessive internal rotation and hyperextension, in addition to cranial drawer, excessive internal rotation or



Figure 10-8 Cranial drawer motion is assessed by placing one forefinger on the patella, one thumb on the lateral fabella, the other forefinger on the tibial tuberosity, and the other thumb on the head of the fibula. An attempt is then made to slide the tibia cranially in relation to the femur, while holding the femur in a steady position.

hyperextension may be helpful in diagnosing injury to the ligament. In some cases, partial drawer may be detected. This may result from a partial cranial cruciate rupture, a torn meniscus that has become wedged between the tibia and femur, or a chronic cranial cruciate ligament rupture with periarticular fibrosis providing partial stability of the stifle. In some cases of partial cranial cruciate ligament rupture, there may be drawer motion in flexion but not extension.

An indirect method of evaluating the integrity of the cranial cruciate ligament is the tibial compression test (Figure 10-9). The stifle is flexed to an approximately 90-135-degree angle. One hand is placed on the stifle with the forefinger resting on the tibial tuberosity. The tarsus is slowly flexed with the other hand while keeping the stifle at a fixed angle. This causes tension in the gastrocnemius muscle and common calcaneal tendon that compresses the femur and tibia. In the absence of the cranial cruciate ligament, cranial movement of the tibial tuberosity in relation to the distal femur is detected.

Injury to the caudal cruciate ligament is rare and is usually due to trauma. Other structures are also commonly injured, such as the cranial cruciate ligament, the medial

collateral ligament, or the medial meniscus. To assess the caudal cruciate ligament, the limb is held as for testing cranial drawer, but the tibia is displaced caudally rather than cranially. Diagnosis of tears of the caudal cruciate ligament can be difficult and requires excellent knowledge of the spatial anatomy of the stifle joint. The integrity of the medial and lateral collateral ligaments are assessed by holding the stifle in an extended position while placing valgus and varus stresses on the joint.

The femur is evaluated next. The quadriceps, biceps femoris, semimembranosus, and semitendinosus muscles are palpated for atrophy, swelling, and pain. Although muscle tears are rare in breeds other than sight hounds, the tensor fascia lata and gracilis muscles should be examined. The shaft of the femur is palpated between muscle bellies, being careful not to confuse bone pain with pain from pinching muscles (Figure 10-10). Bone pain and swelling in the distal femur may indicate osteosarcoma in a large breed, whereas pain in the diaphysis of the bone in a skeletally immature dog may indicate panosteitis.

The coxofemoral, or hip joint is also commonly afflicted with orthopedic conditions. The hip joint should be assessed for luxation in dogs with acute lameness of a rear limb as a result of trauma. To do this, fingers are placed on the tuber sacrale, greater trochanter, and tuber ischium (Figure 10-11). The greater trochanter should be below a line drawn from the tuber ischium to the tuber sacrale; if not, then a hip luxation should be suspected. Next, a thumb is placed in a notch between the greater



Figure 10-9 The tibial compression test is a method to indirectly create cranial drawer motion. To perform this maneuver, the stifle is placed in a slightly extended walking position and firmly maintained in this position. The examiner places a forefinger on the tibial tuberosity and the thumb and remaining fingers on the medial and lateral condyles of the femur. The other hand is used to flex the hock. The forefinger on the tibial tuberosity detects any cranial shift of the tibia in relation to the femur as a result of an insufficient cranial cruciate ligament.



Figure 10-10 The femur is palpated for any pain, inflammation, or swelling, while minimizing excessive compression of the muscles.

trochanter and the tuber ischium, and the femur is externally rotated. With a normal hip, the thumb will be displaced out of the notch as the femur is rotated. With a hip luxation, the thumb is not displaced because the greater trochanter is typically displaced cranially and dorsally and does not rotate caudally to displace the thumb from the



Figure 10-11 The relationships of the dorsal border of the wing of the ilium, the greater trochanter of the femur, and the ischiatic tuberosity are established. A triangle should be formed when these structures are outlined. In this photo, the thumb is on the tuber ischium, the forefinger is on the greater trochanter, and the middle finger is on the dorsal border of the wing of the ilium. If the greater trochanter is in a line with the dorsal wing of the ilium and the ischiatic tuberosity, a craniodorsal luxation of the coxofemoral joint may be present.

notch. Another method of evaluating the possibility of a hip luxation is to measure the length of the affected leg. The affected leg will be shorter with the hip extended and will be longer with the hip flexed as compared with the normal contralateral leg. It should be noted that fractures of the femoral neck and capital femoral physal fractures may give similar physical examination findings.

Next the hip should be placed through a range of motion, including flexion, extension, and rotation of the hip joint to determine if pain, crepitus, or decreased range of motion exist. In most cases, dogs have full hip flexion with little or no pain. Patients with osteoarthritis of the hip have pain with hip extension. Subluxation of the hip commonly occurs in young dogs with hip dysplasia. The Ortolani sign may be performed to determine whether subluxation of the coxofemoral joint is present (Figure 10-12). To perform this test, the dog is placed in lateral or dorsal recumbency. The palm of one hand is placed over the region of the sacrum to support the dog, and the fingers of that hand are placed on the greater trochanter and rim of the acetabulum. The femur is grasped with the other hand, and the femur is adducted and pushed dorsally. The examiner may feel the femur displace from the acetabulum. The femur is then slowly abducted while palpating for a change in the relation of the greater trochanter and acetabulum. If the hip was subluxated, a distinct “clunk” will be felt as the hip spontaneously reduces itself as the femur is abducted. An Ortolani maneuver may be performed in a standing dog by

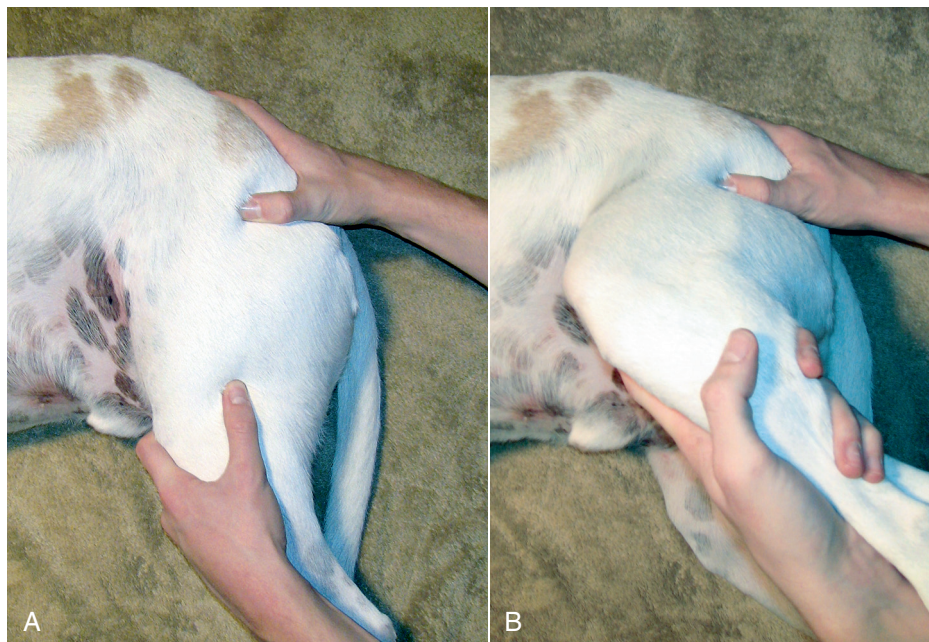


Figure 10-12 A, To perform the Ortolani test, the patient is placed in lateral recumbency, the palm of one hand is placed over the area of the pelvis, the thumb or forefinger of that hand is placed on the greater trochanter, and the femur is adducted while the femur is pushed dorsally with the other hand. B, The femur is then abducted, and if subluxation of the hip has occurred, the reduction will be felt with the thumb or forefinger on the greater trochanter as it is reduced.



Figure 10-13 The phalanges are flexed and extended to determine if any pain or decreased range of motion is present. Here, the digits are flexed as a group.

placing the dog's pelvis and sacrum against the examiner's chest and using one hand to create a dorsal force while adducting the hip.

Forelimbs

Evaluation of the digits, pads, interdigital skin, metacarpal bones, phalangeal joints, and metacarpophalangeal joints is similar to that conducted in the rear limbs (Figure 10-13 and Box 10-3). Careful digital palpation on the dorsal surface of these bones may reveal point pain or tenderness often associated with stress periostitis.

The carpus is evaluated next. The antebrachiocarpal, middle carpal, and carpometacarpal joints are assessed for effusion. The areas of each carpal bone are individually palpated for pain, swelling, and discomfort that may indicate chip fractures or other injuries. The carpus is placed through a complete range of motion to assess for restricted motion, pain, or crepitus. Normally, the carpus should flex so that the pads on the bottom of the foot contact the caudal aspect of the antebrachium. The carpus should extend approximately 10 to 20 degrees beyond vertical (overextension). The collateral ligaments of the carpus are evaluated by placing varus and valgus stresses with the carpus in full extension. There is normally more varus and valgus motion in the carpus than in the other hinge-type joints. The carpus should be placed in a normal walking position and pressure placed on the foot to evaluate for carpal hyperextension syndrome, which is associated with damage to the palmar fibrocartilage and intercarpal ligaments (Figure 10-14). In early hyperextension injuries, there may be pain and reduced flexion of the carpus. The accessory carpal bone should be evaluated for fractures or changes associated with chronic osteoarthritis.

The radius and ulna are assessed for angular limb deformities, and their entire lengths are palpated for pain and

Box 10-3 Common Conditions of the Forelimbs

Common Conditions of the Digits, Pads, and Feet

- Paronychia
- Lacerations of the interdigital skin or foot pads
- Luxations of the interphalangeal joints
- Luxations of the carpometacarpal joint
- Fractures of the metacarpal bones
- Fractures of the phalanges
- Fractures of the sesamoid bones
- Stress periostitis
- Periosteal bone formation associated with hypertrophic osteopathy

Common Conditions of the Carpus

- Fractures of the accessory, radial, ulnar, or numbered carpal bones
- Carpal hyperextension injury
- Flexural contracture of the carpus
- Luxation of the antebrachiocarpal joint
- Collateral ligament injuries
- Stress tendinitis of the ulnar lateralis
- Osteoarthritis

Common Conditions of the Radius, Ulna, and Elbow

- Fragmented coronoid process of the ulna
- Ununited anconeal process
- Osteochondritis dissecans of the medial aspect of the humeral condyle
- Incongruity of the elbow joint
- Fracture of the lateral and medial condyle of the humerus
- Luxation of the elbow joint, usually lateral
- Panosteitis
- Hypertrophic osteodystrophy
- Osteoarthritis
- Fracture of the radius and ulna

Common Conditions of the Humerus

- Fracture of the humerus
- Panosteitis
- Osteosarcoma

Common Problems of the Scapulohumeral Joint

- Avulsion fracture of the supraglenoid tubercle
- Luxation of the scapulohumeral joint
- Fracture of the scapulohumeral joint
- Biceps tenosynovitis
- Medial glenohumeral ligament instability
- Mineralization of the supraspinatus tendon
- Contracture of the infraspinatus muscle
- Osteochondritis dissecans of the caudal head of the humerus
- Osteoarthritis

swelling. The examiner should palpate the medial and lateral styloid processes. Swelling and pain in the metaphyseal region may indicate hypertrophic osteodystrophy in a young, large-breed dog. Pain and swelling in this region in an older dog may indicate the presence of